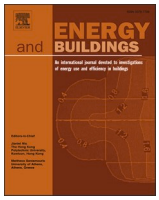
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Risk aware planning of CO_2 transport and geological storage networks for net zero buildings under capacity and injectivity uncertainty

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ABSTRACT

Buildings require deep cuts in both operational and embodied emissions from carbon-intensive materials. Carbon capture, utilization, and storage (CCUS) can support these pathways, but only if CO_2 transport and storage infrastructure remains reliable under uncertainty in pipeline deliverability and reservoir injectivity. This paper develops a network mixed-integer linear programming model for CCUS source-sink design that minimizes discounted net cost while selecting integer pipelines and injection wells. We extend the formulation with a Gamma-robust variant that hedges against per-pipeline capacity shrinkage and per-well injectivity reductions, and a Conditional Value-at-Risk penalty that limits exposure to severe storage shortfalls. The model is evaluated on a fully synthetic dataset enabling controlled comparisons among deterministic, robust, and risk-aware designs. Under representative conditions of moderate transport distances and typical cost structures, hedging a 40% pipeline capacity derate alone increases system cost by 18%, whereas hedging an equivalent injectivity derate increases cost by 52%, and hedging both simultaneously raises cost by 72%. The CVaR analysis further shows that halving the discounted tail shortfall from 200 $MtCO_2$ to 100 $MtCO_2$ requires a 52% increase in expected system cost. These results quantify how uncertainty triggers step changes in well deployment and pipeline sizing, and how risk-aware designs trade higher upfront investment for reduced exposure to extreme shortfalls. Though illustrative, the framework can be instantiated with public data for real regions, providing decision support for CCUS planning linked to net-zero building strategies.

1. Introduction

Buildings sit at the center of the energy and climate agenda because they drive large and persistent demand for electricity and heat and rely on carbon intensive material supply chains that shape both operational and embodied emissions. Global atmospheric CO_2 continues to rise, amplifying climate risks and reinforcing the need for deep mitigation pathways [1]. In many decarbonization roadmaps, Carbon Capture and Storage (CCS) or Carbon Capture, Utilization, and Storage (CCUS) is positioned as a key enabling option for hard to abate parts of the energy system, including large point sources that provide electricity and industrial products linked to the built environment [2,3]. Power generation remains a major contributor to energy related emissions [4], with forecasts for the cement sector under a business-as-usual scenario showing a continued rise in emissions [5]. Meanwhile, the ability to capture, transport, and securely store CO_2 at scale and at acceptable cost will determine whether CCUS can meaningfully support net zero trajectories relevant to buildings and their upstream supply chains [6].

Designing the CO_2 transport and storage system can be formulated as a source sink network problem on a directed graph that connects multiple emission sources to candidate storage sites, optionally through hubs [7,8]. The planner must decide which links to build, how to size pipelines using discrete capacity options, and how many injection wells to deploy at each sink [9]. These investment decisions are coupled with operational planning because annual CO_2 flows must satisfy mass balance relationships while respecting capture limits, pipeline throughput, and injection constraints across the network [10,11]. In this context, network design is not only a cost minimization exercise but also an infrastructure reliability problem, because shortfalls in transport or injection capability directly translate into unmet storage targets. These shortfalls become particularly consequential when examined through the lens of building sector decarbonization.

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Consequently, the reliability of CO_2 transport and storage networks is a direct determinant of the feasibility of net-zero building pathways, as visualized in Fig. 1. A building's net-zero status depends on abating both operational emissions from its energy use and embodied emissions from construction materials such as cement and steel. While CCUS applied to upstream sources such as power plants for electricity and industrial plants for materials enables this abatement, the entire pathway hinges on the consistent performance of the capture, transport, and storage infrastructure. This creates a critical interdependency: uncertainty in pipeline capacity or storage injectivity can reduce CO_2 deliverability and affect the reliability of building-sector decarbonization claims. If this infrastructure underperforms, the expected CO_2 reductions may not be achieved, directly affecting the carbon accounting of net-zero building strategies. Therefore, planning this enabling infrastructure requires a risk-aware framework that explicitly quantifies and mitigates exposure to such deliverability shortfalls.

Despite this critical dependency between reliable infrastructure and building decarbonization, literature on CO_2 source-sink planning remains largely deterministic and therefore can understate exposure to infrastructure underperformance when applied in deployment planning [12]. Two uncertainty sources are particularly consequential for deliverability: effective pipeline capacity can deviate from nominal values due to operational constraints, and storage injectivity can be lower or more variable than expected due to subsurface conditions. Existing approaches that incorporate uncertainty often do so in limited ways, for example via interval style representations or partial hedging, and they rarely quantify the tail risk associated with annual storage shortfalls in a way that is transparent to decision makers [13]. As a result, there remains a gap in practical planning tools that jointly address discrete infrastructure design, bounded capacity and injectivity reductions, and explicit shortfall risk.

This study addresses these gaps by formulating a network Mixed Integer Linear Programing (MILP) for CCUS source to sink design that minimizes discounted net cost while selecting integer sized pipelines and injection wells under standard mass balance and capacity constraints. Building on this core model, a gamma (Γ)-robust variant hedges against per pipeline capacity shrinkage and per well injectivity reductions, and a scenario based CVaR penalty on storage shortfall controls the average of the worst outcomes under uncertainty. The formulations are tested on a fully synthetic yet internally consistent dataset with hypothetical node locations, a complete set of candidate source to sink arcs, and representative techno economic parameters from the literature, enabling controlled comparisons among deterministic, Γ-robust, and CVaR aware designs. While the numerical results are illustrative, the framework can be instantiated for real regions and for built environment relevant decarbonization pathways using public datasets such as the U.S. Environmental Protection Agency's Greenhouse Gas Reporting Program (GHGRP) for plant level emissions [14], the U.S. Energy Information Administration (EIA) for power-plant characteristics [15], and the National Energy Technology Laboratory (NETL) Carbon Storage Atlas for storage candidates [16], together with region specific cost and policy assumptions. The implications of these results for the design of reliable CCUS infrastructure serving net–zero building pathways are examined in detail in the discussion following the presentation of the case study and numerical analysis.

2. Literature review

The pursuit of net-zero building targets has expanded from operational energy use to the full life cycle [17]. Significant progress has been made in reducing operational energy demand through intelligent building controls [18], yet embodied carbon from construction materials like cement and steel remains a critical challenge [19]. Decarbonizing these supply chains is essential, and CCUS is a leading technological pathway for mitigating their process emissions, as demonstrated by recent technoeconomic assessments in the cement sector [20]. However, the contribution of CCUS to building sector decarbonization is not determined by capture technology alone, but also by the guaranteed delivery and storage of CO_2 through dedicated infrastructure networks [21]. The existing literature on CCUS infrastructure planning offers well-established optimization models for designing cost-effective pipeline routes and matching emission sources to storage sites [22]. These models typically assume fixed, known parameters for system capacities.

A defining limitation across these studies is their reliance on deterministic inputs or scenario comparisons, which treat uncertainty as an external sensitivity check rather than an integral design parameter. This approach overlooks the profound technical variability in pipeline hydraulic capacity and geological reservoir injectivity, variability that is well-documented in engineering and geoscience domains. Consequently, infrastructure designs optimized for nominal conditions may be fundamentally vulnerable to underperformance, risking the consistent emission reductions required for credible building sector pathways. Therefore, planning reliable CCUS networks to serve industrial emitters

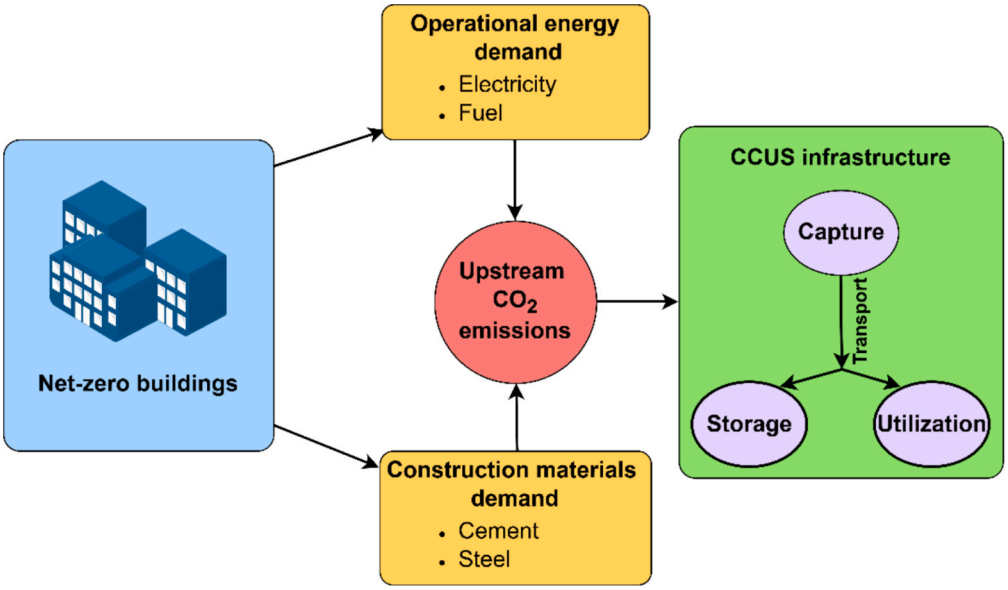


Fig. 1. Interdependence between net-zero building targets and CCUS infrastructure.

requires next-generation models that formally internalize these physical uncertainties and directly quantify the resulting risk to system deliverability.

2.1. Foundational CCUS network modeling and source-sink matching

Planning CO_2 capture, transport, and storage systems has evolved around integrated techno-economic frameworks that represent source-sink networks as costed graphs solved using Mixed integer linear programming (MILP). MILP is widely used in infrastructure planning because it can jointly represent discrete investment decisions within a single optimization model [23–26]. Zhang et al. [27] review how recent CCUS source-sink matching and network models are almost always written as mixed-integer linear programs on costed graphs, often with Geographic Information System (GIS) tools for route generation and clustering, and they note that most of these frameworks still rely on deterministic parameters and scenario analysis rather than formal robustness inside the model. Tan et al. [28] extended this line by proposing a multiperiod source-sink matching formulation that accounts for time, injection-rate limits, and storage capacities, which helps capture temporal feasibility, but they still treat each route as a simple link and do not design shared trunks, hubs, or discrete pipelines and wells in a network. Gu et al. [29] later used a genetic algorithm to jointly select source-sink pairs, routes, and pipeline layouts, gaining flexibility in search but sacrificing the transparency of an exact MILP and omitting detailed discounted economics. These studies show how to match and route CO_2, but none of them simultaneously co-designs hubs, integer pipelines and wells, and finance-aware net present cost in one exact network model. For the capital-intensive planning of shared infrastructure to decarbonize regional industrial clusters, such as those producing steel and cement for the building sector, this gap in integrated, asset-specific design is significant. This study addresses this gap by formulating a network MILP that sizes discrete pipelines and wells, enforces node-wise mass balance, and minimizes discounted net cost for a CCUS cluster.

More recent work moves toward dynamic cluster planning. Wu et al. [30] introduce an optimization-based CCUS cluster model that plans capture, pipeline expansion, and storage deployment over multiple time periods, showing how dynamic targets can change the timing and choice of investments. However, this cluster framework still assumes fixed capacities and does not incorporate uncertainty in pipeline capacity or storage injectivity as a design driver. This represents a critical oversight for infrastructure that must guarantee performance over multi-decadal horizons aligned with building lifespans and decarbonization pledges. In contrast, the present work uses a simpler, synthetic test system but embeds uncertain capacities directly in the network MILP, so that topology, sizing, and operating patterns are optimized under explicit robustness and risk-aversion settings.

2.2. Emerging approaches to uncertainty in CCUS planning

A smaller group of CCUS studies explicitly models uncertainty in planning. He et al. [31] developed both a worst-case MILP for interval uncertainty and a two-stage stochastic MILP for stochastic parameters in source-sink matching, and they showed that robust and stochastic designs can diverge significantly from deterministic solutions. However, their models focus on matching flows between sources and sinks and do not design shared pipelines, hubs, or discrete wells. Wu et al. [32] later proposed an inexact dynamic source-sink matching model for CCUS clusters that combines interval linear programming with chance-constrained programming to plan capture, transport, and storage under both interval and probabilistic uncertainties. That formulation moves beyond simple sensitivity analysis, but it mainly controls the probability of violating injectivity constraints rather than giving planners a clear handle on how many capacities can deteriorate at once or how large the shortfalls might be. For policymakers and investors relying on CCUS to meet sectoral emission budgets for construction materials, understanding not just the probability but also the potential magnitude of a delivery shortfall is essential for effective risk management. Building on these ideas, this work applies a budgeted uncertainty set to both pipeline capacities and well injectivities within an exact network MILP, allowing conservatism to be adjusted as a design parameter.

Another approach explores uncertainty by sampling multiple storage scenarios. Olson and Yaw [33] solve deterministic optimizations for each case and identify a core set of pipelines and wells common to most optimal networks. While insightful, this separates uncertainty from optimization and lacks a coherent risk metric. In contrast, this study integrates uncertainty within the optimization, combining budgeted robustness with Conditional Value-at-Risk on storage shortfall to construct cost-risk tradeoff curves for different risk preferences. This integrated methodology provides a direct link between infrastructure investment and the financial or regulatory risk of failing to sequester promised emissions from the building sector's supply chain.

2.3. Technical underpinnings of uncertainty in transport and storage

The disconnect between deterministic planning models and real-world project risks stems from well-characterized physical uncertainties. The storage and transport literature agrees that injectivity and pipeline capacity are central to CCUS feasibility and inherently uncertain. The Intergovernmental Panel on Climate Change (IPCC) guidelines explain how injectivity depends on reservoir properties, fluid behavior, and well pressures, emphasizing monitoring and pressure management [34]. International Energy Agency Greenhouse Gas R&D Program (IEAGHG) and best-practice documents highlight that early estimates of injectivity and storage capacity can vary widely [35,36], while multiphase-flow simulations show capillary pressure–saturation relationships strongly influence storage efficiency [37]. A recent review notes that capacity estimates for large formations often span broad ranges until more data are collected [38]. These studies underscore the need to treat injectivity and capacity as ranges rather than fixed values. Recent works quantify injectivity and pressure uncertainty for planning models: Mishra et al. [39] estimate injectivity indices from transmissibility data, showing wide plausible intervals within a single formation; Tadjer and Bratvold [40] use Bayesian learning with reservoir ensembles to express storage performance probabilistically; De Simone et al. [41] highlight errors from simple pressure superposition in multiple-sink systems, adding uncertainty to effective injectivity limits. While these studies address injectivity uncertainty, a formal link to source-sink matching (SSM) network design is still lacking. This lack of integration means that sophisticated geoscientific understanding of subsurface risk rarely informs the high-level infrastructure design models used for regional decarbonization planning. This work models injectivity as an uncertain parameter within the network MILP, hedges it below a nominal value, and analyzes its impact on network design and cost.

On the transport side, an IEAGHG technical assessment [42] explains how impurities, phase behavior, and operating pressure change CO_2 density and compression needs, which in turn affect effective pipeline capacity and cost. This results in a design challenge where pipeline capacity is not a fixed value but is variable and complex to determine, as it depends on specific source composition, flow rates, and route topography [43]. Consequently, these inherent technical uncertainties introduce significant gaps into safety planning, where assumptions about failure rates, release models, and safety thresholds critically affect the accuracy of quantitative risk assessments for pipeline infrastructure [44]. Despite this evidence of variable capacity, most existing CCUS network MILPs still assign single deterministic values to this asset. Treating pipeline capacity as fixed ignores a major source of deliverability risk for networks tasked with continuous operation to abate emissions from baseload industrial plants serving the construction

sector. To address this gap, we introduce explicit uncertainty in pipeline capacity by applying a budgeted robust set to shrink available capacities, penalizing the resulting storage shortfalls via a CvaR term in the objective function.

2.4. Risk-averse optimization and its potential for CCUS infrastructure

Given the long-lived and irreversible nature of both building stocks and CO_2 transport pipelines, planning frameworks must evaluate not just average costs but also exposure to high-consequence, low-probability failures. Robust optimization is widely used in energy planning to keep mixed-integer models feasible under parameter uncertainty. Bertsimas and Sim [45] introduced a formulation that limited the number of coefficients that can reach worst-case values, providing a clear tuning parameter for conservatism. Tian et al. [46] applied robust optimization to CO_2 pipelines, showing that uncertainties in temperature, composition, and energy prices can alter diameters and compressor placements. Similar approaches have been used in gas [47] and hydrogen networks [48]. These studies demonstrate that robust sets are practical at scale but focus on gas, hydrogen, or single CO_2 corridors. This work extends robust design to a full CCUS source-sink network, applying it to both pipeline capacities and well injectivities.

Risk-averse optimization complements robust design by explicitly shaping the distribution of outcomes, not just ensuring feasibility. Originating in finance, Value-at-Risk (VaR) sets a probabilistic loss threshold but does not quantify losses beyond it [49]. Rockafellar and Uryasev [50,51] introduced CVaR as a coherent and optimizable measure of tail risk. This approach has been effectively translated to energy infrastructure; for instance, CVaR has been used in joint transmission and generation expansion planning [52] and in hydro scheduling to limit profit variability [53]. These applications demonstrate CVaR's utility in creating explicit cost-risk trade-offs. However, such risk-averse formulations are notably absent in CCUS network design, where uncertainty is still primarily addressed via chance constraints or scenario analysis. This absence is particularly striking because CCUS, like electricity transmission, is a capital-intensive network utility whose failure to perform directly jeopardizes climate targets, in this case, those for embodied carbon in buildings. This work helps fill that gap by incorporating a CVaR term on annual storage shortfall into the CCUS MILP, allowing decision-makers to evaluate how risk aversion influences network topology, asset sizing, and total cost.

Most detailed CCUS source-sink and network studies to date focus primarily on China. Li et al.'s [54] work on the application of CCUS in coal-fired power plants, Sun and Chen's [55] China CCS decision-support system to study carbon-tax impact on national matching, and Nie et al.'s [56] study on combining SSM optimization with the life cycle assessment (LCA) have contributed effectively to the nation, but these studies analyses rely on deterministic parameters and policy scenarios rather than embedding capacity and injectivity uncertainty into the optimization. Though Sun et al. [57] extended these approaches offshore by matching coastal sources to marine sedimentary basins and studying how alternative deployment timings and constraints affect costs and storage potential, but it still is ineffective in handling the uncertainty in the model and the risk averse design. Outside China, CCUS network-planning studies are also emerging in Europe and the United States. Middleton and Bielicki [58] developed the infrastructure model *SimCCS*, which offers a scalable framework for designing CO_2 transport and storage networks and has been applied to potential CCS deployment in the U.S. In Europe, efforts by the Zero Emissions Platform [59] and the European Commission's Joint Research Centre [60] have focused on estimating CO_2 transport and storage costs and identifying potential pipeline corridors, including cross-border CO_2 transport networks shaped by evolving policy and deployment scenarios.

Yet these regional studies remain largely deterministic and scenario-driven, leaving them unable to incorporate uncertainty into optimal source-sink matching, even though they clearly show that CCUS network-design challenges are global and that large-scale models are already used for corridor and cluster planning. For stakeholders seeking to lock in emission reduction pathways for the building sector, this methodological gap translates into an inability to formally weigh the higher upfront cost of a robust network against the risk of infrastructure underperformance and unmet abatement goals. This study addresses these gaps by providing a more generalizable framework that incorporates budgeted shrinkage in pipeline capacity and sink injectivity, evaluates tail risks of unmet capture targets through CVaR, and examines how such hedging strategies affect costs and resource requirements. A synthetic source–sink layout and literature-based techno-economic data are used as a transparent test system for comparing deterministic, robust, and CVaR-aware formulations, which can later be re-parameterized for real regions.

3. Methodology and materials

Fig. 2 presents the comprehensive, three-stage framework developed for risk-aware planning of CO_2 transport and storage infrastructure, which forms the core methodological contribution of this study. The framework is designed to systematically integrate uncertainty into the infrastructure design process, moving from data synthesis through advanced optimization to comparative analysis. The process begins with the data preparation stage, which establishes a harmonized, synthetic dataset to ensure internal consistency for controlled experimentation. This stage involves compiling four cohesive sets of techno-economic parameters. For industrial emission sources, which represent facilities such as cement plants and steel mills that are critical to building material supply chains, key data include annual production, specific emission factors, and the resulting capturable CO_2 volume, paired with associated capture costs. For pipeline corridors, the data encompasses route lengths, nominal transport capacities, and capital and operating costs. For geological storage sinks, the defining parameter is injectivity per well, coupled with site development and well drilling costs. Global system parameters, such as discount rate, project lifetime, and carbon credit profiles, complete the dataset. The use of a synthetic, internally consistent dataset is a deliberate choice to provide a controlled benchmark for comparing deterministic, robust, and risk-averse formulations under identical network and cost assumptions.

The prepared data feeds into the central mathematical modeling - stage, which employs a shared MILP structure to model the source-sink network. This tiered structure is reflected visually in Fig. 2 by the stacked progression from deterministic to robust to risk-averse models within a single optimization pipeline. The foundational model is a deterministic MILP that identifies the cost-optimal network by selecting discrete pipelines and wells under the assumption that all capacity and injectivity parameters are fixed at their nominal values. Building on this, the Gamma-robust variant introduces a proactive hedge against uncertainty. It treats pipeline capacity and well injectivity as parameters that can shrink within a defined band and incorporates a robustness budget. This budget allows the planner to control the degree of conservatism, forcing the design to remain feasible even when multiple components simultaneously degrade toward their worst-case values. The most advanced formulation is the scenario-based CVaR-shortfall model. This model transitions from guaranteeing feasibility to explicitly managing performance risk. It evaluates designs against a set of discrete scenarios representing different realizations of capacity and injectivity. By incorporating a Conditional Value-at-Risk penalty on the annual storage shortfall into the objective function, the model directly discourages designs that would perform poorly in the worst-case scenarios, thereby allowing the optimizer to trace a cost-risk frontier. This progression from deterministic to robust to risk-averse modeling provides a versatile toolkit for planners with different risk preferences.

The final output analysis stage synthesizes and compares the results from the three models to generate actionable planning insights. Solving the deterministic model yields a baseline optimal network topology, its

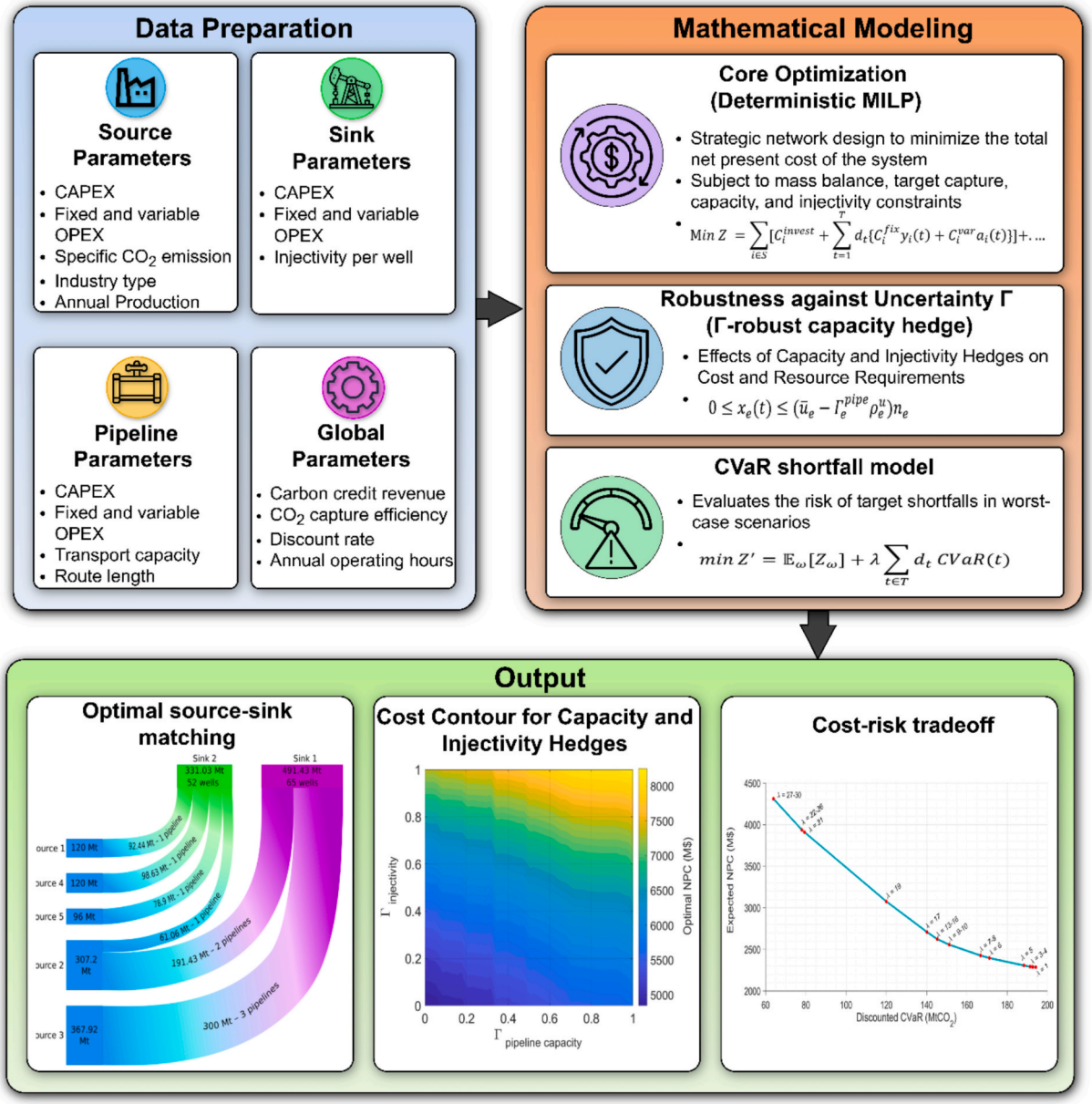


Fig. 2. Overall framework of research.

net present cost, and the associated infrastructure footprint in terms of total pipeline length and number of wells. Executing the robust model across a range of conservatism budgets produces response curves and contour maps that quantify the cost of robustness, revealing how total cost and infrastructure requirements increase as the design is hardened against greater uncertainty. It also identifies critical thresholds where small increases in the robustness budget trigger discrete and significant capital expansions. Solving the CVaR model for different levels of risk aversion generates a series of optimal designs that map out an efficient frontier, plotting the expected system cost against the metric of tail-risk shortfall. Analyzing this frontier shows how much additional investment is required to meaningfully reduce exposure to extreme delivery failures. By comparing outputs across all three formulations, the framework clarifies the trade-offs between nominal cost, robustness to simultaneous capacity shortfalls, and financial risk due to storage target violations, providing a structured basis for decision-making in the face of deep technical uncertainty. The insights generated here provide the empirical basis for the subsequent discussion on implications, limitations, and pathways to real-world application.

3.1. Data preparation

This study compiles techno-economic data for three infrastructure classes in the CO_2 transport-storage system: emission sources, storage sinks, and pipelines. For each industrial source, the dataset includes the facility type, annual production, and the specific CO_2 emission factor (tCO_2 per unit of output), defined as the mass of CO_2 emitted per unit of product and used with annual production to compute annual emissions. These operational data are paired with economics required for the network model: initial capital investment, annual fixed operating cost, variable cost, and end-of-life salvage value. All source-side quantities are harmonized to consistent engineering units prior to modeling.

For each storage sink, the dataset records injectivity, defined as the steady annual storage capacity per well. Sink-side costs comprise of site construction and development cost, capital cost per injection well, annual fixed cost per well, variable cost proportional to injected mass, and salvage value. Following common practice in techno-economic assessments, the annual fixed cost per well is set to 2.5% of well capital cost when not explicitly reported in the source material. These

parameters determine feasible annual injection and the number of wells required to meet the storage target.

Pipeline entries include nominal transport capacity per pipe, route length, and cost parameters: initial investment per kilometer, annual fixed operating cost per kilometer, variable cost proportional to throughput, and salvage value. When not provided directly, the annual fixed cost is taken as 2.5% of pipeline capital expenditure, consistent with the assumption applied on the sink side. Straight-line distances between nodes are used as pipeline lengths unless otherwise stated.

Project-wide parameters complete the dataset: planning horizon, discount rate, carbon-credit revenue schedule, annual capture target as a percentage of capturable CO_2, capture efficiency at sources, and average annual operating hours. All the quantities have been converted to SI units and are consistent. These harmonized inputs ensure internal consistency across sources, sinks, pipelines and support reproducible evaluation of net present cost and infrastructure sizing in the optimization.

All techno-economic data used in this study are therefore representative assumptions rather than site-specific measurements. Capture costs, pipeline capacities, injectivities, and capital expenditure (CAPEX) and operating expenditure (OPEX) values are taken as midpoints of ranges reported in the open literature and are not calibrated to any particular industrial facility, storage formation, or regulatory regime. As a result, the numerical outputs of the model, such as net present cost, pipeline lengths, and well counts, are interpreted as illustrative consequences of these assumed parameters rather than realistic forecasts for an actual CCUS project. The main value of the case study is to demonstrate how the optimization framework responds to changes in capacity and injectivity risk under a coherent but synthetic dataset.

3.2. Mathematical modeling

We formulate a MILP for CCUS source-sink network design over a discrete horizon $t \in T = c(1, \cdots, T)$. The model (i) represents sources, optional hubs, and sinks on a directed graph; (ii) sizes integer assets (parallel pipes on arcs and wells at sinks); (iii) minimizes discounted net cost (CAPEX, OPEX, minus carbon-credit revenue, minus salvage); and (iv) enforces annual mass balance and capacity/injectivity constraints. Two uncertainty treatments are built on the same structure: a Γ-robust capacity hedge and a CVaR-shortfall risk model.

3.2.1. Deterministic MILP model

First, the deterministic MILP model is formulated to design the network and determine the optimal build variables, specifically the number of parallel pipelines and wells, to minimize the NPC of the entire system. Several sets and indices are defined to represent the model structure. Sources are denoted by $i \in S$, sinks by $j \in R$, and hubs by $h \in H$. The complete set of nodes is given by $N = S \cup R \cup H$. Directed pipeline arcs are represented as $e = (m \rightarrow n) \in E \subseteq N \times N$, indicating flow from node m to node n. The planning horizon is divided into discrete years $t \in T = \{1, \ldots, T\}$. For any node k, the notation *out(k)* refers to the set of arcs leaving node k, and *in(k)* refers to the set of arcs entering node k.

The main input parameters for the model are presented below.

Q_i	emissions at source i (t/yr)
$\eta \in (0,1]$	maximum capture fraction at any source
T_{tar}	system-wide capture target (t/yr)
$\overline{u}_e$	nominal capacity of one pipe on arc e (t/yr/pipe)
L_e	pipe length on arc e (km)
$\overline{\tau}_j$	nominal injectivity of one well at sink j (t/yr/well)
C_i^{invest}, C_e^{invest}, C_j^{invest}	initial investment costs for capture ($), pipelines ($/km), and wells ($/well) respectively
C_j^c	One-time construction and development cost of sink j ($)
C_i^{fix}, C_e^{fix}, C_j^{fix}	fixed costs for capture, pipelines, and wells respectively ($/yr)
C_i^{var}, C_e^{var}, C_j^{var}	variable costs for capture, pipelines, and wells respectively ($/t)
S_{ai}, S_{ae}, S_{aj}	salvage values realized at the end of year T for capture, pipelines, and wells ($)

(*continued on next column*)

(*continued*)

R^{rev}	carbon credit revenue per ton stored ($/t)
T	project Lifetime (years)
r	discount rate (%)
$d_t = (1+r)^{-t}$	discount factor

The decision variables in this model are classified into two categories: build variables and operate variables. The build variables are time-invariant integers whose values remain fixed once chosen. They are defined as:

$n_e \in \mathbb{Z}_+$	number of parallel pipes on arc e
$w_j \in \mathbb{Z}_+$	number of wells at sink j

The operating variables are time-varying continuous variables whose values change over time according to different operating conditions. They are defined as:

$y_i(t) = \frac{operatinghoursinyeart}{8760} \in [0,1]$	fraction of time source i operates in year t
$a_i(t) \geq 0$	CO_2 captured at source i in year t (t/yr)
$x_e(t) \geq 0$	flow on arc e in year t (t/yr)
$b_j(t) \geq 0$	CO_2 injected at sink j in year t (t/yr)

The objective of the model is to minimize the total NPC of the CO_2 capture, transport, and storage system. The NPC accounts for the initial investment, annual fixed and variable operating costs, salvage values, and carbon credit revenues over the project lifetime. The formulation is given in Eq. (1).

$$\begin{aligned} MinZ = &\sum_{i\in S}[C_i^{invest} + \sum_{t=1}^{T} d_t\{C_i^{fix} y_i(t) + C_i^{var} a_i(t)\} - d_t S_{ai}] + \sum_{e\in E}[C_e^{invest} L_e n_e \\ &+ \sum_{t=1}^{T} d_t\{C_e^{fix} L_e n_e + C_e^{var} x_e(t)\} - d_t S_{ae}] + \sum_{j\in R}[(C_j^c + C_j^{invest} w_j) \\ &+ \sum_{t=1}^{T} d_t\{C_j^{fix} w_j + C_j^{var} b_j(t) - R^{rev} b_j(t)\} - d_t S_{aj}] \end{aligned} \tag{1}$$

where C_i^{invest}, C_e^{invest}, and C_j^{invest} denote the initial investment costs for capture ($), pipelines ($/km), and wells ($/well), respectively; C_i^{fix}, C_e^{fix}, and C_j^{fix} are the annual fixed operating costs for capture, pipelines, and wells ($/yr); C_i^{var}, C_e^{var}, and C_j^{var} are the variable operating costs for capture, pipelines, and wells ($/t); S_{ai}, S_{ae}, and S_{aj} are the salvage values at the end of year T for capture, pipelines, and wells ($); C_j^c is the one-time construction and development cost of sink j ($); R^{rev} is the carbon credit revenue per ton of CO_2 stored ($/t); n_e is the number of parallel pipes on arc e; w_j is the number of wells at sink j; $a_i(t)$ is the CO_2 captured at source i in year t (t/yr); $x_e(t)$ is the CO_2 flow on arc e in year t (t/yr); $b_j(t)$ is the CO_2 injected at sink j in year t (t/yr); d_t is the discount factor for year t; and $y_i(t)$ is the fraction of time that source i operates in year t.

The model is constrained by a set of physical, operational, and design relationships that ensure feasible CO_2 capture, transport, and storage over the planning horizon. The total amount of CO_2 injected at all sinks must satisfy the system-wide capture target, as expressed in Eq. (2).

$$\sum_{j\in R} b_j(t) \geq T_{tar} \tag{2}$$

The capture at each source is limited by the available emis)sions and the maximum capture fraction η, as shown in Eq. (3):

$$0 \leq a_i(t) \leq \eta Q_i y_i(t), \forall i \in S \tag{3}$$

Mass balance relationships ensure conservation of CO_2 across the network. At each source node, the difference between total outflow and inflow equals the amount captured at that source, as given in Eq. (4):

$$\sum_{e\in out(i)} x_e(t) - \sum_{e\in in(i)} x_e(t) = a_i(t), \forall i \in S \quad (4)$$

Similarly, at each sink node, the total inflow minus outflow equals the CO_2 injected, as expressed in Eq. (5):

$$\sum_{e\in in(j)} x_e(t) - \sum_{e\in out(j)} x_e(t) = b_j(t), \forall j \in R \quad (5)$$

At intermediate hub nodes, mass is conserved such that inflow equals outflow, as defined in Eq. (6):

$$\sum_{e\in in(h)} x_e(t) = \sum_{e\in out(h)} x_e(t), \forall h \in H \quad (6)$$

The transport and storage capacities are restricted by design limits. The CO_2 flow on each pipeline arc cannot exceed its installed capacity, and injection at each sink is bounded by the available number of wells and their nominal injectivity, as formulated in Eq. (7) and Eq. (8):

$$0 \le x_e \le n_e \overline{u}_e, \forall e \in E \quad (7)$$

$$0 \le b_j \le w_j \overline{\tau}_j, \forall j \in R \quad (8)$$

The domain restrictions for all decision variables are defined in Eq. (9):

$$n_e \in \mathbb{Z}^+, w_j \in \mathbb{Z}^+, y_i(t) \in [0,1], a_i(t), x_e(t), b_j(t) \ge 0 \quad (9)$$

3.2.2. Γ-robust capacity hedge

To hedge against possible capacity shortfalls, pipeline and well capacities are treated as uncertain within specified deviation bounds. The nominal pipeline capacity $\overline{u}_e$ and well injectivity $\overline{\tau}_j$ may vary within the ranges.

$$\overline{u}_e - \rho_e^u \le u_e \le \overline{u}_e, \overline{\tau}_j - \rho_j^\tau \le \tau_j \le \overline{\tau}_j$$

where $\rho_e^u \ge 0$ and $\rho_j^\tau \ge 0$ denote the maximum possible deviations in pipeline capacity and well injectivity, respectively

The budget parameters $\Gamma_e^{pipe}, \Gamma_j^{sink} \in [0,1]$ represent the fraction of the worst-case shrinkage enforced in the optimization. A value of $\Gamma = 0$ corresponds to nominal operation, whereas $\Gamma = 1$ enforces the full deviation limits. Incorporating these uncertainty budgets, the pipeline and well-capacity constraints are reformulated as shown in Eq. (10) and Eq. (11).

$$0 \le x_e(t) \le (\overline{u}_e - \Gamma_e^{pipe}\rho_e^u) n_e, \forall e, t \quad (10)$$

$$0 \le b_j(t) \le \left(\overline{\tau}_j - \Gamma_j^{sink}\rho_j^\tau\right) w_j, \forall j, t \quad (11)$$

3.2.3. CVaR-shortfall model

The deterministic and Γ robust models treat capacity uncertainty through worst case shrinkage of pipeline and well capacities but still enforce the annual capture target as a hard constraint. In practice, large undersupply events may occur when both pipeline capacity and reservoir injectivity are weaker than anticipated. To study the cost-risk trade off associated with these events, we extend the network design problem to a two-stage stochastic MILP with a CVaR penalty on annual capture shortfalls.

Uncertainty in the transport-storage system is represented by a finite set of scenarios

$\Omega = \{1, \cdots, \omega\}$ with probabilities p_ω that describe simultaneous derating of all pipelines and wells. In each scenario ω, the realized capacity of one pipe on arc e and the realized injectivity of one well at sink j are denoted by u_e^ω and τ_j^ω. Consistent with the uncertainty band used in the Γ robust analysis, realizations lie within a 40% lower deviation from nominal values.

The build and well decisions (n_e, w_j) are first-stage variables that remain fixed across scenarios, while the operating variables $(a_i^\omega(t), x_e^\omega(t), b_j^\omega(t), y_i^\omega(t))$ are second-stage variables that adapt to each scenario ω. These variables satisfy the same mass balance and capture bound equations as in the deterministic MILP, but with scenario specific capacity constraints shown in Eq. (12).

$$0 \le x_e^\omega(t) \le n_e u_e^\omega, 0 \le b_j^\omega(t) \le w_j \tau_j^\omega, \forall e, j, t, \omega \quad (12)$$

To allow controlled violation of the annual capture target T_{tar}, we introduce nonnegative shortfall variables $g_\omega(t)$ that measure the unmet portion of the target in each scenario and year. The target constraint involving the shortfall variable is given by Eq. (13).

$$\sum_{j\in R} b_j^\omega(t) + g_\omega(t) \ge T_{tar}, g_\omega(t) \ge 0, \forall t, \omega \quad (13)$$

For a given design (n_e, w_j) and operating decisions, let Z_ω denote the discounted net present cost of scenario ω. Z_ω is evaluated with the same cost expression as the deterministic NPC (Eq. (1)) but using the scenario dependent variables $a_i^\omega(t)$, $x_e^\omega(t)$, and $b_j^\omega(t)$. The expected discounted cost is shown in Eq. (14).

$$\mathbb{E}_\omega[Z_\omega] = \sum_{\omega\in\Omega} p_\omega Z_\omega \quad (14)$$

We quantify tail risk of the annual shortfall process with the CVaR measure at confidence level $\alpha \in (0,1)$. For each year t we introduce a VaR variable $\eta(t)$ and nonnegative auxiliary variables $s_\omega(t)$ and impose the standard linear CVaR constraints shown in Eq. (15).

$$s_\omega(t) \ge g_\omega(t) - \eta(t), s_\omega(t) \ge 0, \forall t, \omega \quad (15)$$

The risk associated with shortfalls is not fully described by the mean of $g_\omega(t)$. Large but rare shortfalls are particularly undesirable from a policy and system planning perspective. To focus on these tail events, we use the CVaR of the annual shortfall process at confidence level $\alpha \in (0, 1)$. CVaR at level α measures the expected shortfall in the worst $(1-\alpha)$ fraction of scenarios. The CVaR of the shortfall in year t is obtained by averaging these excesses over all scenarios and scaling by $1/(1-\alpha)$, as shown in Eq. (16).

$$\mathrm{CVaR}(t) = \eta(t) + \frac{1}{1-\alpha}\sum_{\omega\in\Omega} p_\omega s_\omega(t) \quad (16)$$

In this study we mainly use $\alpha = 0.95$, which focuses the risk measure on the most adverse tail of the modeled shortfall outcomes across scenarios. With the finite scenario set used here, this corresponds to emphasizing the most severe capacity/injectivity realizations represented in Ω. The risk-averse model combines expected cost and shortfall risk in a single scalar objective. Let d_t denote the usual discount factor for year t. The total cost functional, denoted by Z', adds a discounted sum of yearly CVaR terms to the expected NPC. The resulting two-stage CVaR formulation is summarized in Eq. (17):

$$minZ' = \mathbb{E}_\omega[Z_\omega] + \lambda \sum_{t\in T} d_t \, CVaR(t) \quad (17)$$

The parameter $\lambda \ge 0$ has units of cost per ton of discounted shortfall and controls the degree of risk aversion. Small values of λ produce designs close to the expected-cost optimum and tolerate larger shortfalls in rare low-capacity years. Larger values of λ incentivize building additional pipelines and wells that hedge against low-capacity scenarios and reduce the high tail of the annual shortfall distribution.

In the numerical experiments, λ is varied from 1 to 50 \$/tCO_2. Solving Eq. (17) for each λ yields one point on the cost risk frontier, characterized by the expected NPC, the discounted CVaR of shortfall, the mean annual shortfall, and the corresponding build decisions such as total pipeline length and total number of wells.

3.3. Output

The framework generates a set of outputs that describe the optimal design, its economic performance, and its response to uncertainty. First, the deterministic model provides the baseline solution: the optimal network topology (selected routes, pipelines per arc, and wells per sink)

along with summary metrics such as the NPC, total pipeline length, and total storage capacity. Second, the Γ-robust analyses generate response curves and contour plots that illustrate how the NPC, total pipeline kilometers, and number of wells vary with changes in the robustness budgets for pipeline capacity and well injectivity. These outputs identify which uncertainty source drives cost most significantly and reveal threshold values of Γ where discrete investments in additional pipelines or wells are triggered. Third, the CVaR-shortfall formulation reports, for a range of risk-aversion coefficients, the expected NPC, the discounted CVaR of annual shortfall, the mean annual shortfall, and the corresponding infrastructure sizing. These results define a cost–risk frontier and show staircase changes in pipeline length and well counts, quantifying the investment needed to reduce exposure to large storage deficits. Together, the outputs from all three formulations enable a direct comparison of design strategies, clarifying the trade-offs between nominal cost, robustness to capacity uncertainty, and financial risk due to storage shortfalls.

4. Case study

The case study employs a stylized carbon capture, transport, and storage (CCTS) network to demonstrate the proposed optimization framework. Fig. 3 illustrates the synthetic layout comprising five emission source (S_1-S_5), one intermediate hub (H_1), and two potential storage sinks (R_1-R_2). The figure is drawn to scale; the scale bar at lower left indicates distances in kilometers. Inter-site distances are measured as straight-line separations between node coordinates, and pipeline lengths in the optimization model are approximated by these straight-line distances. This is a common assumption in early-stage CO_2 pipeline planning, where detailed right-of-way information is unavailable, and the study region is large enough that moderate routing deviations change absolute lengths but not the relative ordering of candidate routes. The locations and labels of all nodes are purely hypothetical and do not correspond to any real industrial plants, hubs, or storage complexes. This is a deliberate choice: the network is intentionally designed as a synthetic test system to exercise the deterministic, Γ-robust, and CVaR-shortfall formulations under controlled conditions. The red nodes (S) represent upstream industrial CO_2 sources, the blue node (H) denotes a hub that can consolidate flows or enable trunk pipeline routing, and the green nodes (R) indicate deep-saline aquifer storage sites. Dashed lines correspond to all candidate pipeline arcs considered in the optimization model, defining feasible source-to-hub-to-sink transport paths. This layout therefore serves as a numerical experiment for evaluating model behavior under consistent network topology and parameterization.

The data for the CO_2 sources are summarized in Tables 1 and 2. Table 1 describes the characteristics of the five sources (S_1-S_5), which are assumed to represent a cement plant, an iron and steel mill, a coal-fired power plant, another cement plant, and an ammonia synthesis plant, respectively, along with their annual production capacities and

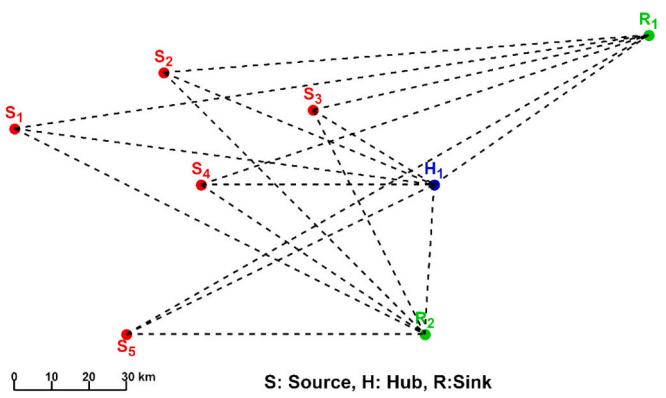


Fig. 3. Synthetic CCUS source-sink-hub network used for analysis.

Table 1
Characteristics of CO_2 emission sources used in the model.

Source	Industrial type	Annual production capacity	Specific CO_2 emission (tCO_2/t product)	Source of specific emission data
S_1	Cement plant	10 Mt cement	0.6 tCO_2/ton cement	[68]
S_2	Iron & steel mill	8 Mt steel	1.92 tCO_2/ton steel	[69]
S_3	Coal-fired power plant	2000 MW	2.31 lb CO_2/kWh	[70]
S_4	Cement plant	10 Mt cement	0.6 tCO_2/ton cement	[68]
S_5	Ammonia synthesis plant	2 Mt ammonia	2.4 tCO_2/ton ammonia	[71]

Note: *Specific emission values are obtained from literature sources shown in the last column, while production capacities are assumed for modeling purposes.*

Table 2
Annual CO_2 emission and associated cost parameters for each industrial source.

Source	Annual emission ($MtCO_2$/year)	Capital cost ($M)	Fixed cost ($M/year)	Variable cost ($/tCO2)	Salvage value ($M)
S_1	6	432	13.095	53.99	21.6
S_2	15.36	1105.92	32.832	47.155	55.296
S_3	18.396	1200.339	35.4307	54.4	60.01695
S_4	6	432	13.095	53.99	21.6
S_5	4.8	82.08	6.6744	10.045	4.104

specific CO_2 emissions per unit of product. The selection of these specific industries is directly connected to net-zero building pathways through operational energy use and embodied materials. The cement plants and the iron & steel mill represent primary sources of embodied carbon in construction materials. The coal-fired power plant represents a significant source of operational carbon from building electricity supply. This composition allows the model to evaluate infrastructure planning for the exact industrial clusters that must be decarbonized to achieve credible net-zero building pathways.

Table 2 summarizes the annual CO_2 emissions, capital cost, fixed operating cost, and variable operating cost for each industrial source. The data are compiled from literature source [32]. In the referenced study, cost parameters are presented as interval ranges; however, the midpoint of each interval is adopted here for consistency. These values are chosen to be broadly representative of large industrial facilities and are used solely to parameterize the synthetic case study. The capital cost, originally expressed in $\$M/tCO_2$, is converted into total capital investment ($M) by multiplying it with the annual quantity of CO_2 captured. Similarly, the normalized fixed operating cost ($\$M/tCO_2$) is multiplied by the annual captured CO_2 to obtain the annual fixed cost ($M/year). Capture efficiency for post-combustion capture systems is typically reported in the range of 85–95% [34]; a uniform value of 90% is applied in this study. The salvage value for chemical and process plants generally ranges from 0 to 10% of the initial capital cost [61], and a representative value of 5% is assumed for a project lifetime of 20 years. The annual CO_2 emissions for each industry ($MtCO_2$/year) are then estimated by multiplying the annual production capacity by the corresponding specific emission factor provided in Table 1.

The data related to the CO_2 storage sinks are presented in Table 3. The values are obtained from the same literature source as the CO_2 source data, which in turn derived the sink parameters following the cost estimates of De Simone and Krevor [41]. Two sinks are considered in this study, all assumed to be DSAs, which represent the most viable and geographically extensive option for large-scale CO_2 storage. As with the sources, these sink parameters are representative of deep-saline storage options reported in the literature and should be viewed as

Table 3
Injectivity and cost parameters of sinks used in the model.

Sink	Injectivity per well (MtCO2/ year)	Sink construction & Development cost ($M)	Capital cost ($M/ well)	Fixed cost ($M/year/ well)	Variable cost ($/tCO2)
R_1	0.3798	29.2635	26.726	0.66815	2.935
R_2	0.3183	26.3375	29.3985	0.734963	2.935

plausible but hypothetical values, not as characterization of a specific formation.

The dataset includes nominal injectivity ($MtCO_2$ $year^{-1}$ $well^{-1}$), sink construction and development cost ($M), capital cost ($M $well^{-1}$), fixed operation cost ($M $year^{-1}$ $well^{-1}$), variable cost ($/$tCO_2$), and salvage value ($M). Nominal injectivity denotes the steady-state capacity of a single well to store CO_2 under specified reservoir conditions. Sink construction and development cost refers to the one-time expenditure required for site characterization, drilling, surface facility setup, and associated infrastructure. Capital cost represents the initial investment for each injection well, while the fixed operation cost covers annual maintenance, monitoring, and administrative expenses. Following literature convention, the annual fixed cost per well is assumed to be 2.5% of its capital cost. The variable cost accounts for operating expenses proportional to the quantity of CO_2 injected, such as compression and site monitoring. The salvage value for each sink is assumed to be zero, reflecting post-closure well abandonment practices [62].

Cost data in the literature were originally reported as intervals with varying confidence levels; in this study, the midpoint values at a 0.05 confidence level were adopted.

Fig. 4 presents a heat map of pairwise straight-line distances for the CO_2 transport network. Rows list origins (H_1 and sources S_1-S_5), while columns list destinations (H_1 and sinks R_1-R_2). Each cell reports the distance in kilometers to two decimals. The accompanying color scale is monotonic: lighter shades represent shorter distances, and progressively darker shades represent longer distances. Distances range from 40.08 km ($H_1 \rightarrow R_2$) to 171.83 km ($S_1 \rightarrow R_1$). Aggregating by source, S_1 has the largest average distance to the sinks, whereas S_3 is the most proximate. These distances are computed from the facility coordinates used in the scaled schematic and are applied directly as pipeline lengths in the transport-cost calculations.

The cost parameters for the pipeline system are derived from the same literature sources used for the capture and storage data. Reported cost ranges in the literature are represented by their midpoint values in this study. The nominal transport capacity of each pipeline is set at 5 Mt CO_2 yr^{-1} per pipe, with an initial investment cost of 1.0899 $M km^{-1}. The variable transport cost is taken as 0.0475 $ tCO_2^{-1} km^{-1}, while the annual fixed cost is estimated as 2.5% of the capital cost, corresponding to an annualized value of 0.02725 $M km^{-1} yr^{-1}. The salvage value of pipelines is assumed to be zero at the end of the project lifetime, consistent with decommissioning guidelines that exclude scrap credits and treat recovery revenues as uncertain; pipelines are typically cleaned, isolated, and abandoned in place [63].

The remaining model parameters are summarized in Table 4. The project lifetime and discount rate are set to 20 years and 10%, respectively, consistent with techno-economic assessments reported by the IEAGHG [64]. The carbon-credit revenue is valued at US $ 85 tCO_2^{-1} for the first 12 years, corresponding to the current U.S. 45Q incentive for geologically sequestered CO_2 [65], and assumed to be zero thereafter. The annual capture target is defined as 90% of the total CO_2 that can be captured, corresponding to 90% of the product of the maximum capture efficiency and total emissions, which reflects a standard design value for post-combustion capture systems [66]. Industrial facilities such as cement plants typically operate for approximately 8000 h per year, corresponding to an availability of about 91% [67]. For consistency across all emission sources, the same annual operating duration of 8000 $hr\text{-}yr^{-1}$ is adopted in this study. Together with the assumed source and sink parameters, these choices define a consistent but hypothetical scenario that allows the proposed optimization formulations to be explored in a transparent way.

5. Results

The results are reported in three stages: (i) a deterministic baseline using nominal inputs, (ii) Γ-robust designs that hedge against uncertainty in pipeline transport capacity and well injectivity, and (iii) a CVaR-shortfall formulation that evaluates tail-risk in meeting the annual storage target under capacity realizations drawn from discrete scenarios. Collectively, these stages compare baseline behavior, the cost and sizing response to robustness, and the cost-risk trade-off under risk aversion.

All results in this section relate to the synthetic network and representative techno-economic parameters described in the case study section. Consequently, the absolute magnitudes of net present cost, pipeline length, well counts, and annual flows are interpreted as illustrative outputs that reveal how the proposed deterministic, Γ-robust, and CVaR-

Table 4
Key economic and operational parameters adopted in the model.

Parameter	Unit	Value
Project lifetime	years	20
Discount rate	%	10
Carbon credit revenue	$/tCO2	85 (for $t \leq 12$)
Annual CO_2 capture target	% of capturable CO_2	90
Annual operating time of sources	hours/yr	8000

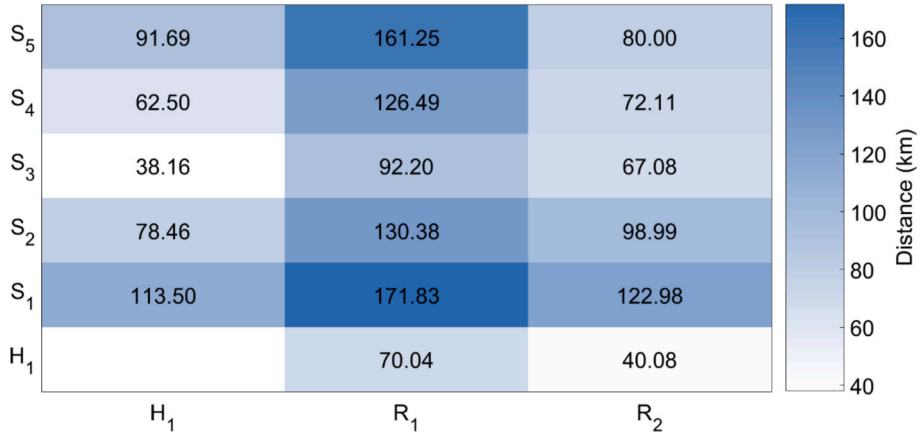


Fig. 4. Pipeline distances for source-sink-hub pairs (km).

shortfall formulations behave under controlled changes in capacity and injectivity risk. The reported values are specific to the synthetic case and are used primarily to compare formulations under consistent assumptions. Accordingly, the results are used to extract qualitative and structural insights, such as the ranking of uncertainty drivers between injectivity and transport, threshold effects from integer capacity additions, and the shape of cost-risk trade-offs, rather than to predict absolute infrastructure requirements in any specific region.

5.1. Baseline deterministic MILP results

The deterministic MILP is first solved using the nominal input data, and the resulting optimized CO_2 transport-storage configuration is shown in Fig. 5. The Sankey diagram summarizes the cumulative CO_2 transfer over the 20-year horizon: emission sources (S_1-S_5) appear on the left, and the two deep-saline aquifer sinks R_1 and R_2 appear at the top. Ribbon widths are proportional to the 20-year cumulative flow on each arc, and labels report the associated flow (Mt) and number of parallel pipelines. The bars at R_1 and R_2 indicate the total CO_2 stored at each sink together with the corresponding number of injection wells sized by the optimization model.

Over the planning horizon, the network captures and stores about 822 Mt of CO_2 out of roughly 1011 Mt emitted by all sources, satisfying the design capture target. The optimized configuration minimizes the total system cost with an NPC of USD 4.8 billion, installing 117 wells and a baseline pipeline system of about 911 km. Because this is a synthetic test system parameterized with representative literature midpoints and assumed geometry, these absolute quantities (USD, km, wells) are not intended as project-level predictions; they serve only as a baseline for comparing how the formulations change topology, sizing, and shortfall risk under uncertainty.

The topology in Fig. 5 exhibits two key features. First, storage is skewed toward R_1: it receives about 491 Mt of CO_2 and 65 wells, whereas R_2 receives about 331 Mt and 52 wells. This skewness arises because the assumed data give R_1 higher injectivity and more favorable transport distances, leading the optimizer to develop a few high-capacity trunk routes into R_1 while using R_2 primarily as a secondary outlet. Second, CO_2 supply is dominated by the two largest sources: S_2 and S_3 together contribute roughly two-thirds of the total stored volume, while S_1, S_4, and S_5 contribute smaller but still significant amounts.

Policy incentives also shape the operating pattern. Although Fig. 5 displays only cumulative flows, inspection of the annual flow profiles shows a modest reduction in flow on some arcs after year 12, for example on connections from S_1 and S_2. This change is driven by the expiration of the 45Q carbon credit at year 12, which lowers the marginal benefit of CO_2 delivery in later years and leads the optimizer to reduce throughput slightly while keeping total lifetime storage and infrastructure sizing unchanged. Mass balance is satisfied at each node in all years: captured CO_2 equals net outflow at sources, and net inflow equals injection at sinks. Under nominal capacities, the deterministic solution meets the annual target and provides the baseline for the Γ-robust and CVaR-shortfall analyses.

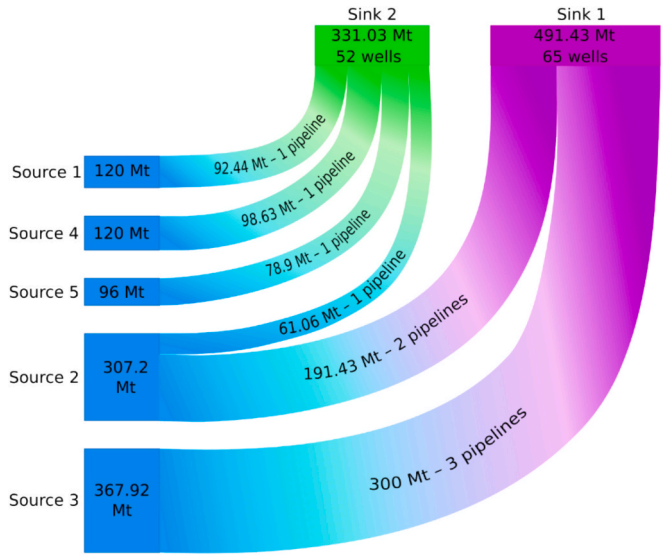


Fig. 5. Deterministic MILP results for CO_2 transport and storage system.

5.2. Robustness to capacity uncertainty: NPC, pipelines, and wells

To quantify the impact of parameter uncertainty, a Γ-robust formulation is applied to pipeline transport capacity and storage-well injectivity. The analysis treats pipeline capacity per pipe and well injectivity per well as uncertain within a symmetric deviation band of 40% below nominal. The robustness budget $\Gamma \in [0, 1]$ specifies the fraction of that worst-case shortfall that the optimizer must hedge. A value of $\Gamma = 0$ recovers the deterministic baseline, while $\Gamma = 1$ enforces the full 40% derate. The study evaluates how Γ affects three outcomes: the optimal NPC, the total installed pipeline-kilometers, and the total number of wells. To separate effects, three sweeps are examined: $\Gamma_{\text{pipeline capacity}}$ varies with $\Gamma_{\text{injectivity}} = 0$; $\Gamma_{\text{injectivity}}$ varies with $\Gamma_{\text{pipeline capacity}} = 0$; and a diagonal sweep sets $\Gamma_{\text{pipeline capacity}} = \Gamma_{\text{injectivity}}$ over the unit interval.

Fig. 6 reports the NPC response under three sweep designs, with markedly different sensitivities. The optimal NPC increases monotonically with Γ in all cases. The deterministic baseline at $\Gamma = 0$ is about USD 4.8 billion. When only pipeline capacity is hedged, the green curve rises gently to about USD 5.8 billion at $\Gamma = 1$, which is an increase of 18.9% relative to the baseline. This comparatively mild slope indicates that the NPC increases only gradually as the pipeline-capacity hedge Γ grows. The network still meets the target with limited additions of parallel lines and minor routing changes. In effect, the transport layer provides enough flexibility that modest derates in per-pipe capacity are absorbed without large capital expansion.

Uncertainty in injectivity produces a much steeper cost response. With $\Gamma_{\text{injectivity}}$ varying and $\Gamma_{\text{pipeline capacity}} = 0$, the red curve grows from the same baseline to about USD 7.3 billion at $\Gamma = 1$. That level is about 51.8% above the baseline and about 27.7% above the pipeline-only hedge at the same Γ. This pattern follows from the role of injectivity in the system: reducing injectivity limits how much each well can store in a year, so the optimizer adds wells and associated site infrastructure to maintain the annual target. Because these additions are discrete and capital intensive, they account for a large share of the NPC increase as Γ grows.

The simultaneous hedge, shown by the blue curve where $\Gamma_{\text{pipeline capacity}} = \Gamma_{\text{injectivity}}$, forms the upper envelope. As Γ increases, the optimizer must protect against concurrent reductions in per-pipe capacity and per-well injectivity, which compresses slack on both the transport and storage sides of the network. At $\Gamma = 1$ the optimal NPC rises to approximately USD 8.3 billion, reflecting the compounded need for additional parallel pipelines together with expanded well capacity and associated site infrastructure.

Taken together, the results indicate that uncertainty in well injectivity is the dominant driver of cost risk in this system, while pipeline-capacity uncertainty has a secondary effect. Across the tested range, the NPC increase is larger for injectivity hedging than for pipeline-capacity hedging, and the combined hedge yields the highest NPC.

Fig. 7 presents the corresponding contour map of NPC over the two-dimensional robustness space, with Γ for pipeline capacity on the horizontal axis and Γ for well injectivity on the vertical axis. Both parameters vary independently from 0 to 1, and color indicates NPC in $M. This surface therefore shows the joint effect on NPC when Γ for pipeline capacity and Γ for well injectivity vary independently.

The surface rises smoothly from the lower-left corner toward the

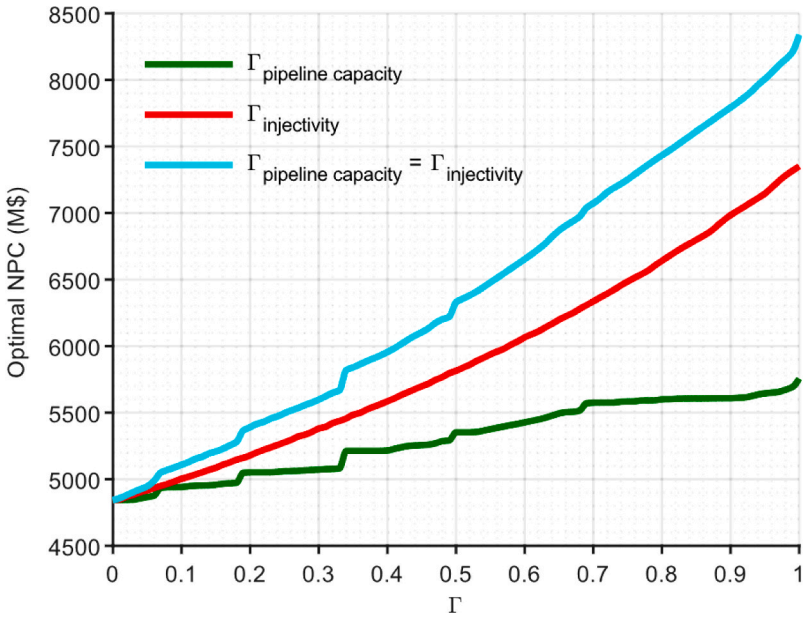


Fig. 6. Optimal NPC under Γ-robust hedges in pipeline capacity and well injectivity.

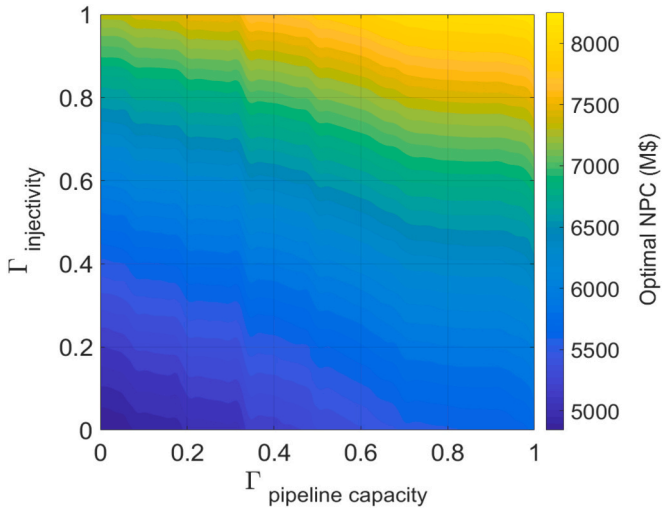


Fig. 7. NPC contour over Γ for pipeline capacity and Γ for well injectivity.

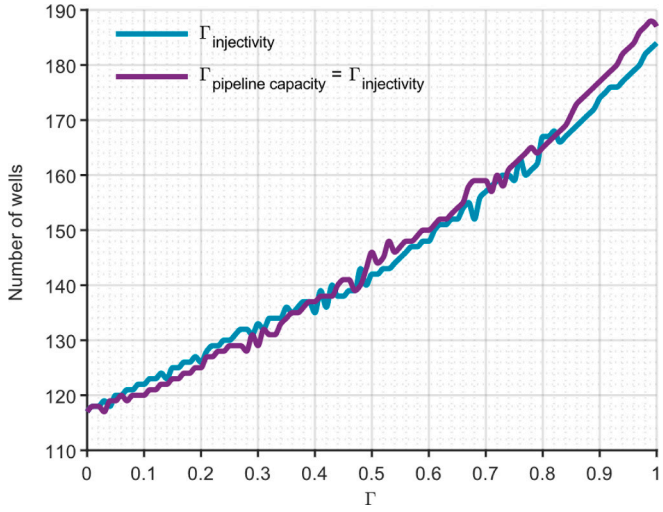


Fig. 8. Effect of robustness budget Γ on the required number of wells.

upper-right, confirming that stronger hedging on either capacity component increases total cost. Moving vertically, by increasing $\Gamma_{\text{injectivity}}$ at fixed $\Gamma_{\text{pipeline capacity}}$, produces larger changes in NPC than moving horizontally by the same amount. This again indicates that injectivity uncertainty is the dominant cost driver in this system. The contour lines are approximately diagonal, which suggests that the cost contributions of the two hedges are close to additive. Faint bands and kinks across the plot correspond to integer expansions in pipelines and wells: as Γ passes certain thresholds, new capacity modules are added, creating visible ridges in the cost surface. This contour map therefore provides a compact way to read trade-offs: for any cost ceiling, the contour of that level identifies the combinations of $\Gamma_{\text{pipeline capacity}}$ and $\Gamma_{\text{injectivity}}$ that keep NPC at or below the threshold, and the tighter spacing in the vertical direction confirms the dominant influence of injectivity hedging on project cost.

Fig. 8 reports how the required number of wells varies with the robustness budget Γ under two settings: (i) Γ applied only to well injectivity with Γ for pipeline capacity fixed at zero, and (ii) Γ applied equally to both injectivity and pipeline capacity. The pipeline-only sweep is omitted because it leaves the well count essentially unchanged across the range, which confirms that transport derates do not materially determine the sizing of storage wells in this network.

When Γ affects only injectivity, the well count increases in an almost linear fashion from 117 wells at $\Gamma = 0$ to about 184 wells at $\Gamma = 1$. This pattern follows from the injectivity derate applied in the constraint. When Γ is applied to both injectivity and pipeline capacity, the total rises slightly further to about 187 wells at $\Gamma = 1$. The two curves are nearly indistinguishable for small Γ and begin to separate only beyond $\Gamma = 0.4$, where pipeline derates interact with reduced injectivity and a few additional wells are drilled to avoid new bottlenecks. The small gap between the curves even at high Γ shows that injectivity uncertainty is the primary determinant of well requirements, while pipeline-capacity uncertainty has only a secondary influence on the number of wells.

Fig. 9 shows how installed pipeline length responds to the robustness budget Γ. As in Fig. 8, two settings are reported: Γ applied only to pipeline capacity with the injectivity hedge fixed at zero (brown curve), and Γ applied equally to pipeline capacity and well injectivity (magenta curve). The injectivity-only sweep is omitted because uncertainty in

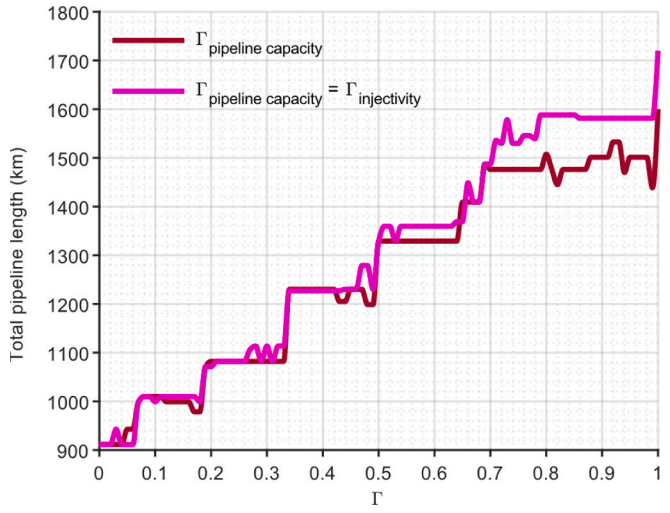


Fig. 9. Effect of robustness budget Γ on the total pipeline length.

injectivity alone has little to no effect on total pipeline length.

The deterministic design at $\Gamma = 0$ installs about 911 km of pipe. As Γ for pipeline capacity increases, total length follows a staircase pattern: flat segments correspond to ranges where the existing topology remains optimal, and upward jumps mark the addition of at least one new parallel line on a bottleneck arc. Under the pipeline-only hedge, pipeline length reaches about 1600 km at $\Gamma = 1$. When injectivity is hedged at the same time, the staircase pattern persists but shifts upward and some steps occur at smaller Γ values. The system then reaches roughly 1720 km at $\Gamma = 1$. This amplification occurs because hedging injectivity increases the number of wells and can shift flow allocation between sinks, which in turn requires extra line capacity on trunk routes. Together, the two curves show that pipeline-capacity uncertainty alone already drives substantial expansions in line-kilometers, and that adding injectivity uncertainty pushes the system toward slightly longer networks and earlier topology changes.

5.3. CVaR shortfall optimization and cost-risk trade offs

The CVaR formulation extends the deterministic and Γ-robust analyses by asking how much additional investment is justified to reduce the risk of large annual storage shortfalls under weak capacity conditions. Rather than enforcing the annual capture target in every realization, the stochastic MILP allows controlled violations and penalizes them in the objective function. This reflects a planning attitude where small, infrequent shortfalls are acceptable, but large misses in bad years should be strongly discouraged.

Uncertainty in pipeline capacity and well injectivity is represented by a finite scenario set. Five scenarios are constructed by applying global derating factors of 1.0, 0.9, 0.8, 0.7 and 0.6 to all pipes and wells, and each scenario is assigned an equal probability. For any given design and operating strategy, the model evaluates the discounted net present cost in each scenario together with the annual shortfall of the capture target. The confidence level is fixed at $\alpha = 0.95$. At this level, CVaR(t) represents the expected annual shortfall conditional on being in the worst 5% of years, so the metric focuses on low-probability but high-impact storage deficits. To make this risk measure comparable with the economic objective, the year-specific CVaR(t) values are discounted with the same factors d_t used in the NPC calculation and summed over the 20-year horizon. Risk aversion is controlled by the coefficient λ in the objective function, which multiplies the discounted CVaR term. This parameter has units of cost per ton of discounted shortfall ($\$\ tCO_2^{-1}$) and determines how much the optimizer is willing to pay, in expectation, to reduce tail risk. Solving the MILP for a range of λ values yields a set of optimal designs that form a cost-risk frontier in the space of (discounted CVaR, expected NPC).

Fig. 10 plots this frontier, with discounted CVaR on the x-axis and expected NPC on the y-axis. Each labeled point corresponds to an optimal design at a particular λ, and the connecting curve highlights the trade off between economic efficiency and shortfall risk. The curve slopes downward and is mildly convex: as discounted CVaR decreases, expected NPC increases, implying that a higher cost is required to reduce the risk. For small λ, the optimization is dominated by the expected-cost term, giving designs with low NPC (about USD 2.3 billion) but high discounted CVaR (around 194 $MtCO_2$). These configurations rely on relatively modest infrastructure and therefore remain vulnerable to very large shortfalls when pipeline and well capacities are strongly derated. As λ increases, the points move along the curve toward lower CVaR and higher NPC because the model invests in additional pipelines and wells that are seldom fully used in average years but reduce shortfalls in low-capacity scenarios. For larger λ, the frontier bends and then gradually flattens. In the middle portion of the curve, moderate increases in NPC buy substantial reductions in discounted CVaR, reflecting the addition of a few key capacity expansions that significantly improve performance in adverse years. Near the left-hand end of the frontier, discounted CVaR falls to roughly 60–70 $MtCO_2$, but expected NPC rises to well above USD 4 billion, indicating that most cost-effective risk-reduction options have already been exploited and further reductions in tail risk require disproportionately large investments. Therefore, the shape of the frontier illustrates a general pattern, though these numerical values are specific to the synthetic network: stronger protection against large shortfalls always comes at a higher expected cost. In a real planning context, such a frontier can help decision makers identify regions where additional spending still yields meaningful reductions in discounted CVaR, and where further risk reduction becomes disproportionately expensive.

Fig. 11 shows how the mean annual capture shortfall varies with the risk-aversion coefficient λ, averaged over 20 years and the five equal-probability derating scenarios. For small λ values, the shortfall remains high, around 23–25 Mt yr^{-1}. In this regime the solution is effectively risk neutral: tolerating recurrent shortfalls is less costly than expanding transport and injection capacity, so the optimization closely resembles the cost-focused stochastic baseline. As λ increases, the curve drops in several distinct steps instead of falling smoothly. Each step indicates that, beyond a certain penalty level, it becomes optimal to add a new block of capacity, such as more wells or an extra trunk line, which significantly reduces the average shortfall. Between these steps, raising λ has little effect because the network structure does not change. At higher λ values, the curve flattens and the mean shortfall stabilizes near 15 Mt yr^{-1}. In this region the system already includes substantial spare capacity under most scenarios, and extra investment mainly trims the most extreme years without altering the long-term average very much. Although the precise λ values at which the steps occur are specific to this case study, the figure illustrates three regimes: a cost-focused regime with high mean shortfall, a transition regime where capacity is expanded in discrete increments, and a saturation regime where additional penalties on shortfalls give only marginal reductions in the mean value.

Fig. 12 shows how the optimal number of wells changes as the risk-aversion coefficient λ increases. The curve has a staircase form: the well count remains constant over intervals of λ and increases in discrete jumps at specific thresholds. Each jump reflects the addition of an integer block of wells that reduces tail shortfalls in the most severe capacity-derating scenarios. For low λ, the solution uses roughly 105–110 wells, which is close to the configuration obtained when only expected cost is minimized. As λ increases through certain thresholds, the well count jumps first into the 120–130 range, then toward about 150 wells, and finally to 186 wells at the highest λ values considered. These jumps line up with the major drops in mean shortfall in Fig. 11. This indicates that, in this synthetic network, the main mechanism for reducing shortfall risk is to increase injectivity by drilling new wells and

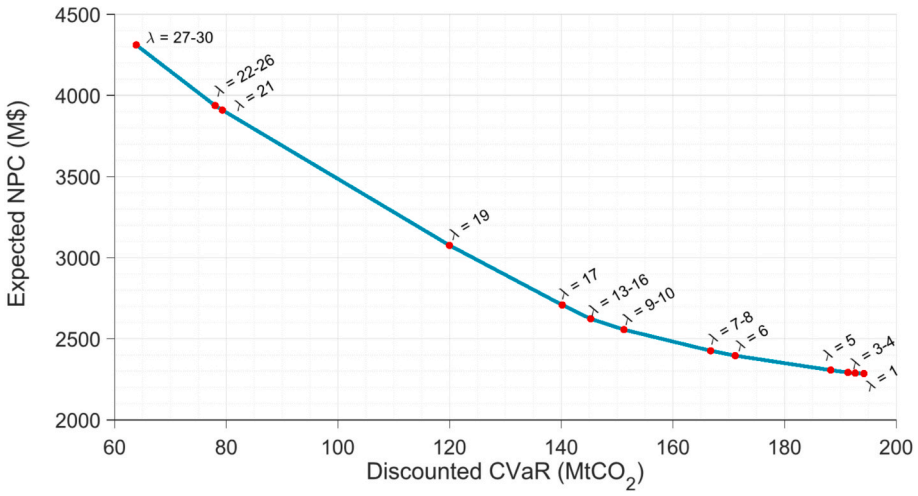


Fig. 10. Cost-risk frontier under the CVaR shortfall formulation (λ in $/tCO$_2$).

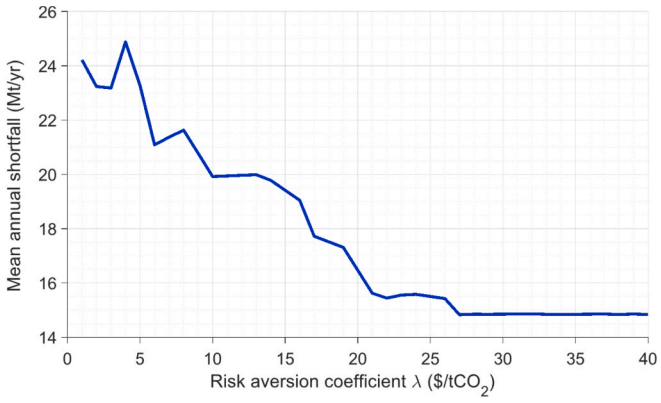


Fig. 11. Effect of risk aversion coefficient λ on mean annual shortfall.

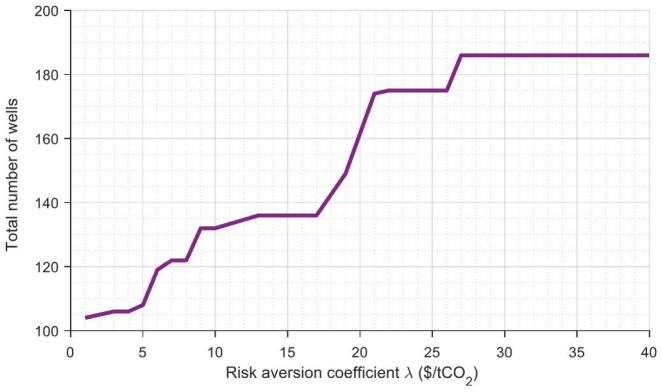


Fig. 12. Total number of wells vs. risk aversion coefficient λ.

developing additional storage capacity, rather than by making small adjustments to operating schedules alone. The absolute numbers are case-specific, but the staircase shape is informative: risk aversion translates into discrete expansions in storage infrastructure at identifiable λ thresholds. The staircase pattern thus highlights three design regimes: a low-λ region where the model accepts frequent shortfalls and keeps the well count near the deterministic level; a middle region where successive well expansions produce substantial reductions in risk; and a high-λ region where further increases in risk aversion raise the implicit cost of shortfalls without justifying additional drilling.

Fig. 13 shows the effect of λ on total installed pipeline length. Here too the curve is piecewise constant with sharp upward jumps, reflecting the indivisible nature of pipeline investments. For small λ, total pipeline length is about 820 km. As λ increases and shortfalls become more heavily penalized, the model begins to add parallel lines along key trunk routes to preserve deliverability under low-capacity scenarios. This creates successive plateaus near 1000 km, about 1080 km, and approximately 1214 km. At higher λ values, the length increases further to around 1512 km and then to about 1636 km, after which no additional expansions occur within the tested λ range. Taken together with the well-count results, this staircase pattern shows that risk-averse designs reduce shortfall exposure primarily by adding capacity in discrete increments rather than through gradual adjustments. Small changes in λ within a plateau leave the network unchanged, while crossing a critical λ threshold triggers the emergence of a new bottleneck and requires an additional pipeline expansion to further reduce discounted CVaR. These threshold points therefore mark settings where marginal increases in risk aversion imply substantial additional pipeline investment.

6. Discussion

The findings in Section 5 show that accounting for uncertainty can fundamentally change both the design and the economics of CO_2 transport and storage networks. The following discussion interprets those results to clarify the framework's methodological contribution, explain what the observed design shifts imply for reliable decarbonization of the built environment through electricity and construction-material supply chains, assess how sensitive the conclusions are to key

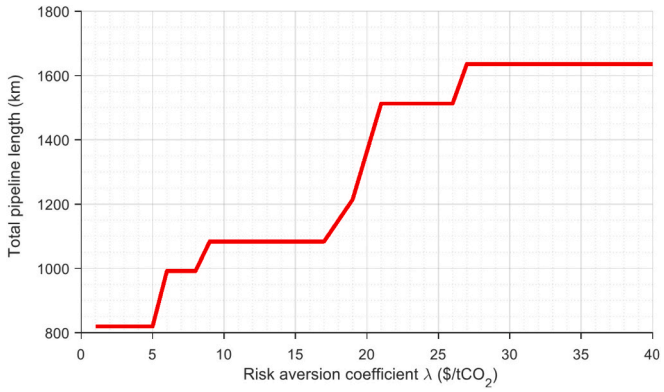


Fig. 13. Effect of risk aversion coefficient λ on total pipeline length.

assumptions, acknowledge limitations for real-world deployment, and outline how the proposed risk-aware planning approach can be applied in regional net-zero building and infrastructure strategies.

6.1. Methodological contributions and design rationale

The primary contribution of this work lies in its integrated modeling framework, purpose-built to internalize two pervasive forms of technical uncertainty into the early-stage planning of CCUS networks. This study addresses a central planning question of how the enabling transport and storage infrastructure must be designed to remain deliverable when key components underperform, given that CCUS is required to support net-zero building pathways. To illustrate this effect, the case study employs a synthetic but internally consistent dataset. This choice demonstrates the execution of the framework and shows how uncertainties in capacities and injectivities affect the optimal infrastructure and its associated cost. It also quantifies the risk associated with these uncertainties under representative techno-economic conditions and industry operations responsible for handling the operational and construction demands of buildings. However, in real regions, outputs depend on many location-specific factors, such as rights-of-way, permitting, routing constraints, formation-specific injectivity, and local cost conditions. Consequently, the results should be interpreted as qualitative and structural insights rather than numerical forecasts. Accordingly, absolute costs, pipeline lengths, and well counts reported in Section 5 are used only to support relative comparisons (deterministic vs. Γ-robust vs. CVaR) and to reveal qualitative design trade-offs. So, the results are interpreted as transferable structural insights rather than region-specific numerical forecasts. In other words, the framework demonstrates how uncertainty changes optimal topology, sizing, and performance risk in ways expected to persist when the model is re-parameterized with real data.

A key methodological insight from the results is that CCUS network expansion under uncertainty is typically not gradual. Because pipelines and wells are installed as integer assets, the optimal design often remains unchanged over a range of uncertainty settings and then shifts suddenly when a bottleneck is reached. This step-like behavior is not a numerical artifact, but a direct consequence of discrete investment decisions combined with binding capacity and injectivity constraints. For planning, this is significant because small increases in conservatism can trigger large increments in capital investment once the system crosses a critical point. The framework therefore helps identify breakpoints where robustness or risk aversion translates into discrete infrastructure additions. This information is crucial when evaluating whether reliability targets justify the next block of spending.

The results also provide insight into how the framework separates and compares different approaches for handling uncertainty. The Γ-robust model operationalizes clear planning decisions regarding how much capacity degradation the network should tolerate and how many components might degrade at once. It clearly demonstrates which uncertain parameter dominantly drives system cost and associated risk. By contrast, the CVaR-shortfall model answers a different but equally practical question regarding the additional cost a planner must incur to reduce exposure to rare but severe storage shortfalls. These choices lead to different design responses in the results. The Γ-robust formulation hardens the system against bounded underperformance by tightening effective capacities, while the CVaR formulation allows shortfalls but discourages designs that perform poorly in the worst scenarios. Presenting both methods within the same network MILP structure is useful. It shows how reliability goals can be expressed either as conservatism against degradation, through robustness budgets, or as aversion to extreme under-delivery, through tail-risk penalties, without changing the basic physical accounting of the CCUS system.

6.2. Implications for net-zero building pathways, infrastructure planning, and policy

The results in Section 5, while based on a synthetic case, highlight a core planning issue for net-zero building pathways: CCUS-supported mitigation for electricity and construction materials requires delivery assurance, not only least-cost feasibility. The following subsections interpret the results in terms of infrastructure planning and policy.

6.2.1. Uncompensated risk in nominally optimal networks

The deterministic optimal solution presented in Fig. 5 establishes a reference infrastructure network designed under the assumption of perfect technical performance. This baseline system meets the prescribed annual capture target at the lowest net present cost under the assumption of perfect technical performance. While the specific network topology is dependent on the synthetic case study parameters, the broader implication for net-zero building strategies is universally relevant. Decarbonization claims for both building operational energy and embodied construction materials fundamentally depend on the premise that captured CO_2 is reliably sequestered annually, not merely on average over time.

This deterministic result also provides a clear illustration of how policy and economic incentives can influence operational decisions within a fixed infrastructure system. The modeled reduction in CO_2 flows after year twelve corresponds directly to the expiration of the carbon credit incentive. This illustrates that time-limited incentives can change utilization even when infrastructure is fixed, which weakens the durability of net-zero accounting if delivery is not governed.

Most significantly, the deterministic baseline represents a system highly vulnerable to underperformance. As demonstrated in Fig. 6 through Fig. 9, introducing uncertainty in pipeline capacity or storage injectivity necessitates substantial and discrete investments in additional infrastructure to maintain the same annual capture target. Translated to the context of building sector decarbonization, this reveals that the credibility of a net-zero pathway does not hinge solely on the theoretical availability of CCUS technology, but critically on the engineered reliability of the transport and storage infrastructure under realistic deviations from ideal performance assumptions.

6.2.2. Injectivity uncertainty as the dominant cost driver

The analysis reveals that, under the assumed techno-economic conditions and network geography, the credibility of net-zero building pathways is disproportionately vulnerable to uncertainty in geological storage performance compared to transport infrastructure. As shown in Fig. 6, enforcing a worst-case hedge on pipeline capacity alone increases the optimal net present cost by approximately 18% relative to the deterministic baseline. By contrast, hedging against injectivity uncertainty alone increases cost by approximately 52%, while hedging against both uncertainties raises cost by approximately 72%. Although case-specific, the ranking identifies which uncertainty most threatens deliverable CO_2 storage that underpins operational and embodied mitigation. In the context of this study, which assumes typical pipeline and storage cost structures and a relatively simple geography with shorter transport distances, injectivity uncertainty emerges as the dominant driver of cost escalation. This is because the system must compensate for injectivity risk by adding discrete storage capacity to maintain annual deliverability. However, in more complex geographies or constrained permitting environments where routing is difficult and transport distances are long, pipeline additions can become disproportionately expensive. Under such conditions, transport-capacity uncertainty may become comparable to, or even exceed, injectivity uncertainty in its impact on total system cost.

The primary mechanism is that injectivity is a direct annual

throughput limit at the sink, and the network has few substitutes when that limit tightens. If injectivity is lower than expected, each well can store less CO_2 per year, so the only reliable way to maintain an annual capture and storage target is to add storage capacity, most commonly by drilling additional wells. This response is visible in Fig. 8 through the sharp increase in well counts as the injectivity robustness budget increases. Because wells and site development are discrete and capital-intensive, storage underperformance translates into a steep cost response and directly threatens annual delivery that underpins net-zero accounting for both operational and embodied emissions.

This result is consistent with storage practice: injectivity is uncertain during early screening because it depends on subsurface heterogeneity and pressure evolution, and effective rates are further constrained by pressure and plume-management requirements intended to protect caprock integrity and manage induced-seismicity risk. When realized injectivity falls below planning values, corrective actions typically require additional wells or alternative sinks, which increases cost and can delay delivery.

Pipeline-capacity uncertainty is structured differently and can be more flexible to manage within a network. Transport derates usually create bottlenecks on specific arcs, and the system can often respond by adding parallel lines on those routes. Fig. 9 shows that pipeline kilometers can increase substantially as the pipeline-capacity robustness budget rises. However, under the assumed cost structure and relatively short transport distances, those additions increase total net present cost less than the storage-side expansions required under injectivity uncertainty, as reflected in Fig. 6. Fig. 7 reinforces the same interpretation: cost increases more rapidly with injectivity hedging than with pipeline-capacity hedging, and the ridges in the contour surface indicate threshold effects where a small increase in conservatism triggers an integer jump in wells or pipes. Under conditions similar to those assumed here, the results support early emphasis on storage-side characterization and contingency capacity, while noting that long distances, routing constraints, and difficult terrain can shift cost sensitivity toward transport.

6.2.3. Translating tail risk into a performance metric

The CVaR formulation operationalizes the concept of infrastructure reliability by quantifying exposure to severe delivery shortfalls. For a confidence level of 0.95, the CVaR represents the expected shortfall conditioned on the worst 5% of modeled outcomes. This metric directly addresses a central concern for net-zero building pathways: the risk that significant, though infrequent, infrastructure underperformance undermines the annual carbon accounting for embodied materials and clean energy. A discounted CVaR of 100 $MtCO_2$, for example, indicates that the present-value, horizon-aggregated tail shortfall is on the order of 100 million tons under the most adverse portion of the modeled outcome distribution. This reflects a discounted and cumulative measure of risk across the full planning horizon, rather than an annual shortfall or a statement about the worst 5% of individual years. This frames reliability not in terms of average delivery but in terms of resilience against consequential failures.

Interpreted against climate targets that are tracked through annual compliance and cumulative carbon budgets, a worst-tail outcome corresponds to rare but consequential under-delivery years in which credited sequestration may fall short, potentially requiring make-up abatement, credit recovery, and contingency injections or alternate sinks to avoid budget overshoot. The CVaR metric thus quantifies not only physical shortfall but also the regulatory and financial exposure embedded in infrastructure design.

Fig. 10 translates this risk measure into an explicit cost-risk frontier, illustrating the financial investment required to reduce tail-risk exposure. Under the studied conditions, reducing the discounted CVaR from approximately 200 $MtCO_2$ to 100 $MtCO_2$ necessitates a 52% increase in system cost. This quantifies the premium for enhanced delivery assurance, enabling policymakers to balance economic constraints against the need for reliable abatement from building-sector supply chains. Such a frontier allows a regulatory body to set a CVaR-based reliability standard, for instance, a maximum allowable tail-risk exposure, and immediately understand the associated infrastructure investment mandate.

Fig. 11 through Fig. 13 explain how this reliability is purchased in infrastructure terms. Fig. 11 shows step-like reductions in mean annual shortfall as risk aversion increases, indicating that risk reduction is driven mainly by discrete changes in build decisions rather than gradual operational tuning. Figs. 12 and 13 show the corresponding step increases in wells and pipeline length. The planning implication is that tightening reliability requirements can trigger discontinuous capital expansions. For net-zero building strategies, this supports staged deployment and contingency design, such as pre-permitting additional well pads, securing sink expansion options, and protecting transport corridors for upgrades, because modest changes in reliability targets can require the next capacity block.

This framing also responds directly to policy needs. First, CVaR can be used to define reliability standards in performance terms rather than prescriptive design rules. Programs that credit CCUS toward compliance or net-zero claims can require that proposed infrastructure demonstrate a CVaR below a specified threshold, thereby limiting exposure to severe under-delivery under plausible concurrent derating of transport and injectivity. Second, it supports contingency planning by identifying how much tail shortfall remains at a given design point and where discrete expansions become cost-effective; monitoring data can then be linked to predefined triggers for sink-side or transport-side upgrades. Interpreted this way, the risk-aversion parameter acts as an internal penalty for under-delivery, similar to regulatory penalties, loss of credits, or contractual performance requirements, while CVaR summarizes the shortfalls that most strongly affect the credibility of net-zero building pathways.

6.2.4. Policy-relevant implications for delivery-assured infrastructure

Collectively, sections 6.2.1 through 6.2.3 show that CCUS-supported net-zero building pathways depend on delivery assurance that is measurable and governable, not only on least-cost feasibility.

A primary implication is that infrastructure reliability can be treated as a measurable and enforceable performance standard. The cost-risk frontier in Fig. 10 provides a direct mechanism for operationalizing this principle by quantifying the investment premium required to reduce exposure to severe shortfalls. Policymakers can use this relationship to define and justify reliability criteria for infrastructure approval and carbon-crediting programs, and to differentiate between average expected abatement and delivery-assured abatement that supports durable net-zero building claims.

The analysis also indicates that reliability improvements often require discrete, non-marginal capital expansions rather than incremental tuning. The stepwise capacity changes in Fig. 9, Fig. 12, and Fig. 13 indicate that modest changes in performance standards, risk tolerance, or technical assessments can trigger threshold increases in required investment and permitting scope. Net-zero building strategies that depend on CCUS therefore benefit from adaptive, phased deployment plans that anticipate these breakpoints, including pre-permitted sink expansion options, secured pipeline corridors that preserve upgrade feasibility, and monitoring protocols linked to predefined contingency actions.

An essential, context-dependent insight under the representative cost structures and moderate transport distances analyzed is the leverage of storage-side risk management. Under these conditions, investments that reduce uncertainty in storage deliverability or increase sink flexibility can provide larger risk reduction per dollar than transport overbuild, because annual sequestration feasibility is directly governed by sink throughput and its constraints. This supports a practical sequencing for planning: first de-risk storage deliverability through characterization and injectivity testing, pressure-management planning, and sink optionality, then right-size transport to connect sources to verified,

deliverable sinks. At the same time, the framework is designed to test whether this hierarchy holds in a given region. In settings with long transport distances, difficult terrain, limited routing options, or stringent corridor permitting, transport-side uncertainty and the cost of capacity expansions may become comparably consequential for both cost and reliability.

Ultimately, this work provides a replicable template for moving from generic infrastructure advocacy to targeted, risk-informed investment that supports building-sector decarbonization. By instantiating the model with local data on industrial emissions, storage resources, routing constraints, regulatory limits, and costs, planners can identify the binding constraints for a specific net-zero building pathway and justify investments that are not only economically motivated but demonstrably resilient. By quantifying the trade-off between upfront cost and delivery assurance, the framework provides an evidentiary basis for policies and investment strategies that safeguard the long-run integrity of net-zero building goals under real-world uncertainty.

6.3. Sensitivity of conclusions to policy and regional assumptions

This study is designed to extract structural insights from a controlled case, but several conclusions depend on assumptions about incentives, geography, and demand. The discussion below clarifies which findings are expected to persist across settings and which are likely to shift when those assumptions change.

6.3.1. Sensitivity to carbon-credit level, duration, and policy persistence

The results assume an 85 $/t$CO_2$ credit available for 12 years, which affects both utilization and optimal sizing. The deterministic solution already reflects this influence through the post year 12 reduction in CO_2 flows on selected arcs, even though the installed infrastructure remains fixed. This indicates that time-limited incentives can change how a network is operated and how much of the installed capacity is actually used, which matters for the durability of net-zero building claims that rely on stable upstream abatement.

If the credit value is lower, or the credit duration is shorter, the model generally has less economic margin to justify larger networks, redundant capacity, or high levels of robustness. In such settings, the least-cost solution is more likely to concentrate on the lowest-cost source to sink connections, defer expansions that only pay off in adverse conditions, and accept a higher risk of under-delivery in bad realizations. Conversely, if the credit value is higher or persists for longer, additional capacity and higher reliability become easier to justify economically, and robustness or tail-risk reduction can be purchased at lower relative cost because more years of incentivized delivery are available to recover capital investment.

Uncertainty in long-run policy support also interacts with infrastructure risk. A network that is designed to be highly reliable under technical underperformance may still underdeliver climate benefits if incentives weaken or end earlier than expected, because utilization can decline even when capacity exists. This interaction motivates treating policy support as another uncertain driver of delivery, alongside injectivity and transport capacity. The framework can accommodate such alternatives by modifying the time profile and structure of incentives within the objective function. Examples include declining credits over time, credits tied to verified storage performance with time-varying rates, contracts for difference that provide a revenue floor relative to a carbon price, and performance-based payments and penalties linked to annual delivery. In each case, the same physical constraints and investment decisions are preserved, while the economic signals that govern operation and expansion are changed.

6.3.2. Sensitivity to geography, regional storage conditions, and demand profiles

Several behaviors are expected to be robust across regions. The step-like changes in NPC, wells, and pipelines follow from integer investment decisions, so threshold effects should persist whenever capacity is added in discrete blocks. The broader conclusion that deterministic designs can appear cost-efficient while embedding under-delivery risk is also expected to persist, because it follows from optimizing under nominal conditions without explicit protection against underperformance. The CVaR frontier remains broadly transferable because it converts reliability into an explicit cost-risk trade-off that can be used to set performance standards.

By contrast, the relative dominance of injectivity versus transport uncertainty is more context dependent. The injectivity dominance observed here is most likely under typical cost structures with short to moderate distances and relatively unconstrained routing, because storage throughput is a binding annual limit and is difficult to substitute when realized injectivity is below expectation. In regions with long transport distances, difficult terrain, limited corridor availability, or stringent permitting, pipeline additions can become disproportionately expensive, and transport-capacity uncertainty can become comparable to, or exceed, injectivity uncertainty in its effect on cost and deliverability.

Demand-profile assumptions also affect which constraints bind first. If captured CO_2 volumes are relatively steady over time, the design is driven mainly by annual storage throughput, and the key sensitivities remain on the sink side. If volumes ramp up, decline, or shift spatially across sources, the optimal solution becomes more sensitive to transport bottlenecks and to the timing of discrete capacity expansions. Such shifts can arise from power-sector operating patterns and electrification trajectories, as well as from construction cycles that change materials output and associated capture volumes, which in turn change when delivery-assured storage must be available and where expansions are required.

6.4. Limitations, scalability, and upscaling pathway for regional planning

The numerical experiments are illustrative because the case study uses a synthetic but internally consistent dataset and a stylized network representation. The formulation captures capacity and injectivity limits and discrete infrastructure investment, but it abstracts away several real-world features that can materially affect design. These include detailed hydraulics and pressure-flow behavior along pipelines, routing and right-of-way constraints, permitting timelines, site development constraints, CO_2 quality and impurity specifications, and operational dynamics such as ramping, outages, and maintenance. As a result, the numerical magnitudes of cost and infrastructure sizing should not be interpreted as forecasts. The primary value is the structural insight into how uncertainty reshapes topology, sizing, and deliverability risk under controlled conditions.

Future work should strengthen external validity by instantiating the framework for real regions using public data on industrial emissions, candidate storage formations, and feasible transport corridors. Several extensions would improve physical and institutional fidelity while preserving the planning purpose of the model. These include multi-period operation with phased build-out and retirement decisions, richer storage representations that link injectivity to pressure management and regulatory pressure constraints, and improved uncertainty structures that reflect correlation across wells, learning from appraisal data, and scenario generation grounded in site characterization. Additional realism can also be introduced through CO_2 quality constraints, compressor and recompression requirements, and operational constraints that represent contractual delivery obligations and downtime.

Scalability is a substantive challenge for large, real-world systems. The underlying problem is a mixed-integer linear program in which the number of candidate links, investment binaries, and flow variables grows rapidly as the number of sources and sinks increases. The CVaR formulation further increases model size because scenario-based representations expand the variable and constraint sets approximately in proportion to the number of scenarios, while preserving integer

investment decisions that couple all scenarios. Multi-period build-out compounds this growth by adding another dimension to both investment and operation variables. Consequently, national-scale instances with many sources, sinks, candidate corridors, time periods, and scenarios can become computationally demanding, and direct solution with off-the-shelf solvers may not be practical without additional structure.

For regional and national applications, decomposition and tailored approximations are likely to be needed. A practical pathway is to reduce the candidate network before optimization using GIS-based corridor screening, routing feasibility filters, and clustering of nearby sources and sinks to limit the number of arcs and binaries. On the uncertainty side, scenario reduction and structured uncertainty sets can preserve the dominant risk patterns while keeping the model tractable. Algorithmically, variants of Benders decomposition can separate investment decisions from operational and scenario subproblems, while column-and-constraint generation can be used for robust counterparts and large scenario sets. Heuristic or hybrid strategies are also appropriate for early-stage planning, for example solving a coarse aggregated model to identify priority corridors and sinks, then refining locally with higher-fidelity candidate sets and uncertainty descriptions. These approaches allow the framework to remain usable as the scope expands, while retaining the core objective of delivery-assured infrastructure planning.

Despite the above limitations, the framework provides a clear template for linking net-zero building objectives to enabling CCUS infrastructure under uncertainty. A regional application can proceed by translating building-driven decarbonization pathways into sectoral capture demand trajectories for electricity generation and construction-material supply chains, then jointly optimizing transport and storage investments under explicit deliverability risk metrics. The deterministic baseline can be used to identify least-cost corridors and sinks, the robust analysis to test deliverability under bounded underperformance, and the CVaR analysis to quantify the cost of meeting reliability requirements that can be stated as performance standards. This structure also supports phased deployment by identifying discrete expansion thresholds and by linking monitoring outcomes to predefined upgrade actions.

7. Conclusion

This study developed a risk-aware planning framework for CO_2 transport and geological storage networks that supports CCUS-enabled net-zero building pathways by explicitly linking infrastructure design to delivery assurance. A network-based mixed-integer linear programming model was formulated to co-design transport and storage infrastructure by selecting discrete pipeline capacity through parallel pipes and discrete injection capacity through wells, subject to annual mass balance and capacity and injectivity constraints. Two uncertainty treatments were integrated: a Γ-robust hedge that protects feasibility under bounded reductions in per-pipe capacity and per-well injectivity, and a scenario-based two-stage formulation that penalizes annual storage-target shortfall using Conditional Value at Risk at a confidence level of 0.95.

Under representative conditions of moderate transport distances and typical cost structures in the synthetic case study, ensuring deliverability against worst-case injectivity underperformance increases system cost by 52%, nearly three times the 18% increase required to hedge an equivalent pipeline capacity shortfall, while joint worst-case degradation increases cost by 72%. This asymmetry identifies storage injectivity uncertainty as the dominant driver of cost and reliability risk under the assumed conditions. The Conditional Value at Risk results further quantify the price of delivery assurance: halving the discounted tail shortfall from approximately 200 $MtCO_2$ to approximately 100 $MtCO_2$ requires an approximately 52% increase in expected system cost, revealing an explicit cost-risk frontier and threshold-driven expansions in optimal pipelines and wells.

Beyond methodological contribution, the findings have direct implications for policymakers, planners, and infrastructure investors supporting net-zero building strategies. The results indicate that credible CCUS-based abatement claims depend on delivery-assured performance rather than nominal capacity, and that reliability standards expressed through robustness budgets or tail-risk limits can be translated into concrete infrastructure requirements and cost premiums. This enables policy and investment design that explicitly trades upfront spending against the risk of under-delivery that can undermine operational and embodied carbon accounting for buildings.

The numerical magnitudes are illustrative because the case study is synthetic and abstracts routing, permitting, detailed hydraulics, and formation-specific pressure management. Future work should parameterize the framework for real regions using public source and storage datasets, incorporate phased build-out, and represent storage performance uncertainty with pressure constraints, learning, and correlated risks.

CRediT authorship contribution statement

Gaurav Kattel: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Anup Paudel:** Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Seungkeun Yeom:** Writing – review & editing, Methodology, Conceptualization. **Juui Kim:** Writing – review & editing, Methodology, Conceptualization. **Taehoon Hong:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Jun-Ki Choi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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